Organic Chemistry, 2nd Edition

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Chapter 3

An Introduction to Organic Compounds

Nomenclature, Physical Properties, and Representation of Structure

Introduction to organic compounds

Hydrocarbons

(contains only **carbon and hydrogen**)

Saturated hydrocarbon
contain only C-H single bond
Alkanes

Straight-chain alkanes

Branched alkanes

Unsaturated hydrocarbon
contain double and triple bonds
Alkenes and Alkynes

Alkanes

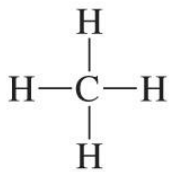
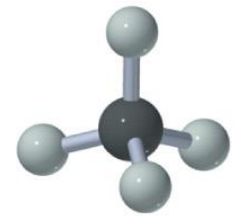
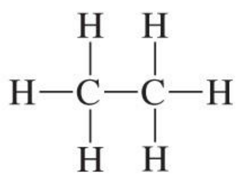
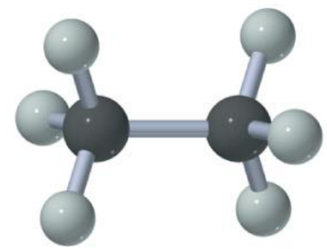
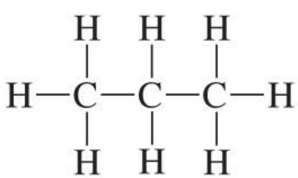
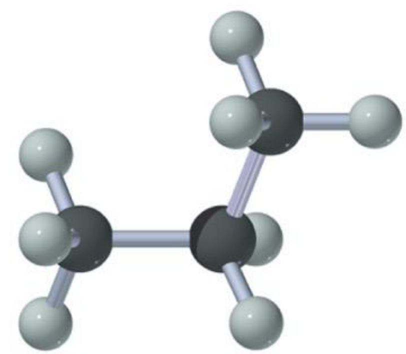
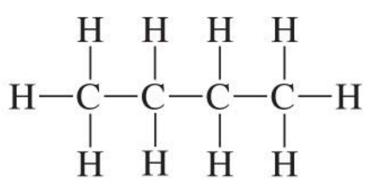
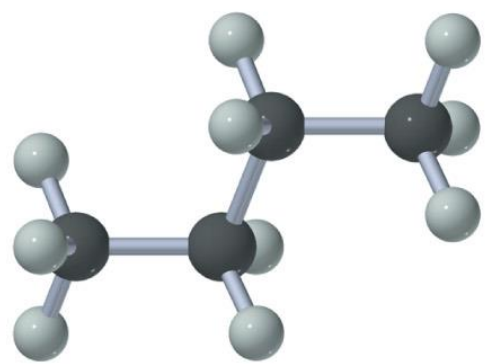
- **Contains only *sp*^{3} hybridized Carbon**
- **only has C-H and C-C single bonds**
  - **Have a general formula of C_nH_{2n+2}**

C Atoms	H Atoms	Formula
1	2 (1) + 2 = 4	CH_4
3	2 (3) + 2 = 8	C_3H_8
6	2 (6) + 2 = 14	C_6H_{14}

the First Ten Continuous-Chain Alkanes

Alkane	Formula
• Methane	CH_4
• Ethane	CH_3CH_3
• Propane	$CH_3CH_2CH_3$
• Butane	$CH_3(CH_2)_2CH_3$
• Pentane	$CH_3(CH_2)_3CH_3$
• Hexane	$CH_3(CH_2)_4CH_3$
• Heptane	$CH_3(CH_2)_5CH_3$
• Octane	$CH_3(CH_2)_6CH_3$
• Nonane	$CH_3(CH_2)_7CH_3$
• Decane	$CH_3(CH_2)_8CH_3$

Memorize

name	Kekulé structure	condensed structure	ball-and-stick model	
methane		CH_4		
ethane		CH_3CH_3		C_2H_6
propane		$CH_3CH_2CH_3$		C_3H_8
butane		$CH_3CH_2CH_2CH_3$		C_4H_{10}

$$\begin{array}{ll} CH_3CH & \!\!- \\ | & \\ CH_3 & \end{array}$$

an “iso”

$$CH_3CH_2CH_2CH_3$$

butane

$$\begin{array}{l} CH_3CHCH_3 \\ | \\ CH_3 \end{array}$$

isobutane

$$CH_3CH_2CH_2CH_2CH_3$$

pentane

$$\begin{array}{l} CH_3CHCH_2CH_3 \\ | \\ CH_3 \end{array}$$

isopentane

$$CH_3CH_2CH_2CH_2CH_2CH_3$$

hexane

hexane

$$\begin{array}{l} CH_3CHCH_2CH_2CH_3 \\ | \\ CH_3 \end{array}$$

isohexane

Alkyl group

$$\text{Alkane} \xrightarrow{-H} \text{Alkyl group}$$

Examples:

$$\text{Meth}\underline{\text{ane}} \xrightarrow{-H} \text{Meth}\underline{\text{yl}}$$

$$CH_4 \xrightarrow{-H} \sim CH_3$$

$$CH_3CH_3 \xrightarrow{-H} CH_3CH_2\sim \quad \text{ethyl}$$

$$CH_3CH_2CH_3 \xrightarrow{-H} CH_3CH_2CH_2\sim \quad \text{Propyl}$$

Alkyl substituents

R—OH an alcohol	CH_3-OH	Methyl alcohol
R—NH_2 an amine	CH_3-NH_2	Methyl amine
R—X an alkyl halide X = F, Cl, Br, or I	CH_3-Cl	Methyl Chloride
R—O—R an ether	CH_3-O-CH_3	Dimethyl ether

Alkyl substituents

$CH_3CH_2NH_2$
ethylamine

$CH_3CH_2CH_2Br$
propyl bromide

$CH_3CH_2CH_2CH_2Cl$
butyl chloride

$CH_3CH_2CH_2NH_2$
propylamine

CH_3I
methyl iodide

$CH_3CH_2OCH_3$
ethyl methyl ether

CH_3CH_2OH
ethyl alcohol

The use of an “iso” prefix:

$CH_3\underset{\substack{| \\ CH_3}}{CH}CH_2CH_2OH$

isopentyl alcohol

$CH_3\underset{\substack{| \\ CH_3}}{CH}CH_2CH_2CH_2Cl$

isohexyl chloride

$CH_3\underset{\substack{| \\ CH_3}}{CH}CH_2NH_2$

isobutylamine

$CH_3\underset{\substack{| \\ CH_3}}{CH}CH_2Br$

isobutyl bromide

$CH_3\underset{\substack{| \\ CH_3}}{CH}CH_2CH_2OH$

isopentyl alcohol

$CH_3\underset{\substack{| \\ CH_3}}{CH}Br$

isopropyl bromide

Table 2.2 Names of Some Common Alkyl Groups

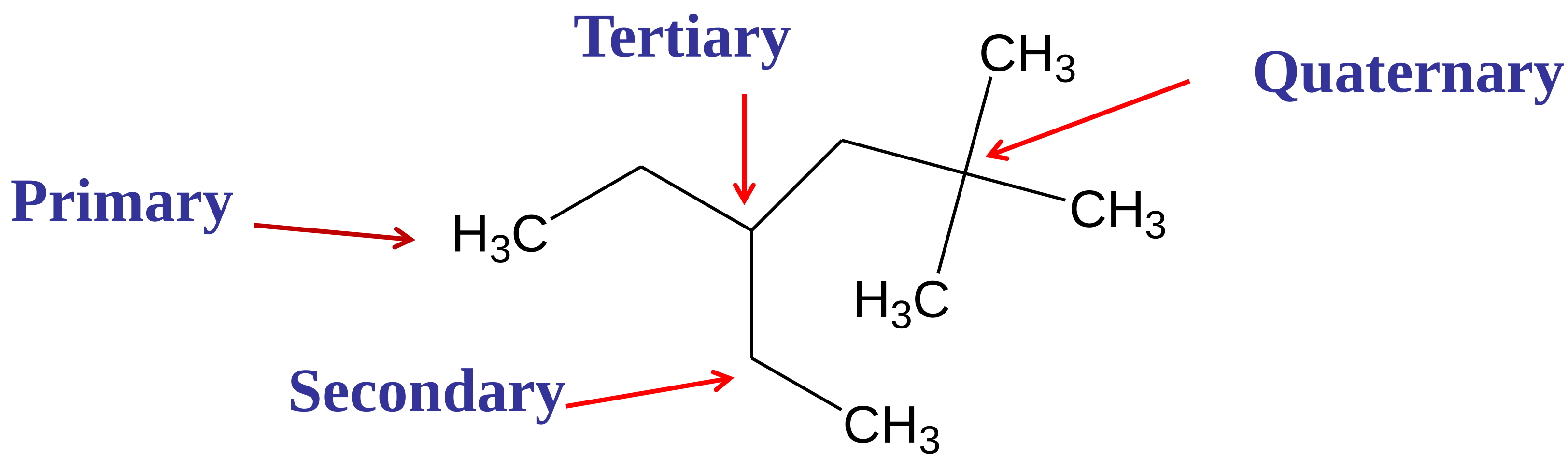
methyl	$CH_3—$	isobutyl	$CH_3\underset{\substack{| \\ CH_3}}{CH}CH_2—$	pentyl	$CH_3CH_2CH_2CH_2CH_2—$
ethyl	$CH_3CH_2—$			isopentyl	$CH_3\underset{\substack{| \\ CH_3}}{CH}CH_2CH_2—$
propyl	$CH_3CH_2CH_2—$	*sec*-butyl	$CH_3CH_2\underset{\substack{| \\ CH_3}}{CH}—$		
isopropyl	$CH_3\underset{\substack{| \\ CH_3}}{CH}—$			hexyl	$CH_3CH_2CH_2CH_2CH_2CH_2—$
butyl	$CH_3CH_2CH_2CH_2—$	*tert*-butyl	$CH_3\overset{\substack{CH_3 \\ |}}{\underset{\substack{| \\ CH_3}}{C}}—$	isohexyl	$CH_3\underset{\substack{| \\ CH_3}}{CH}CH_2CH_2CH_2—$

Primary Carbon is bonded to only one carbon atom

Secondary carbon is bonded to only two carbon atoms

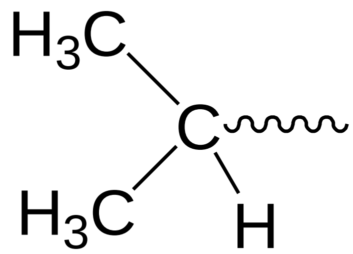
Tertiary carbon is bonded to only three carbon atoms

Quaternary carbon is bonded to only four carbon atoms



Commonly used alkyl groups

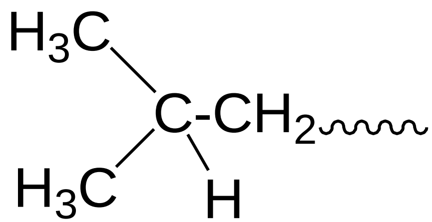
1) Isopropyl



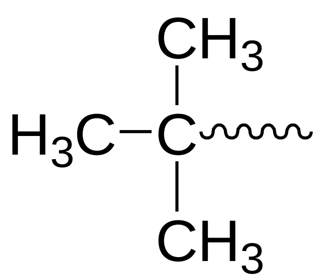
C***H***$_3$CH$_2$C***H***$_3$

Propane

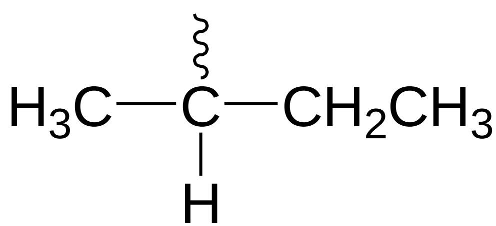
2) Isobutyl



3) tert-butyl



4) sec-butyl



Each has four carbons

from

C***H***$_3$CH$_2$CH$_2$C***H***$_3$

Butane

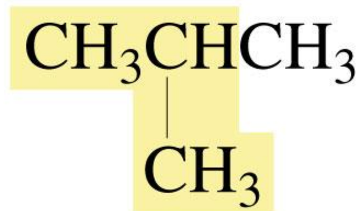
tert = tertiary, sec = secondary

Isomers:

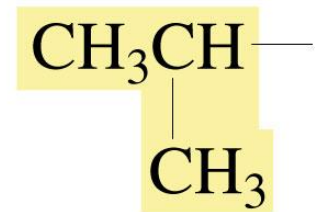
1. Constitutional (Structural) isomers have the same molecular formula, but their atoms are linked differently

$CH_3CH_2CH_2CH_3$

butane



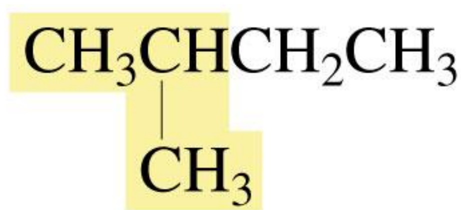
isobutane



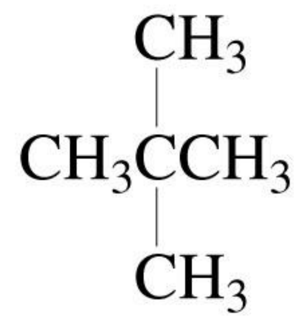
an "iso" structural unit

$CH_3CH_2CH_2CH_2CH_3$

pentane

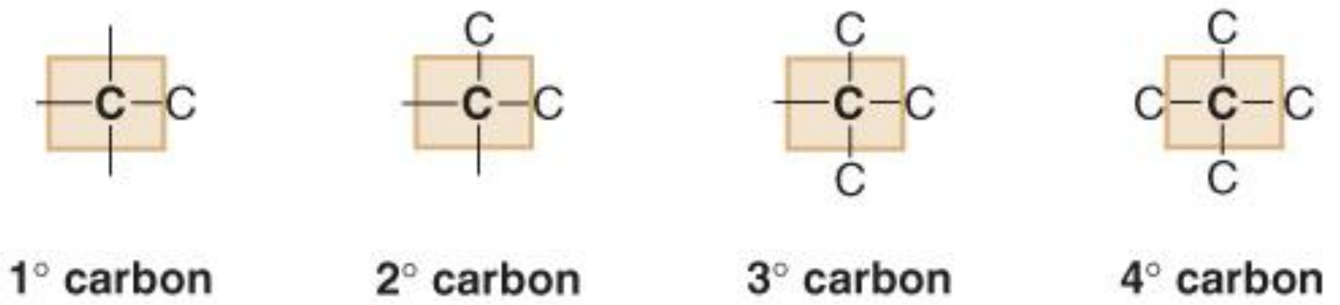


isopentane

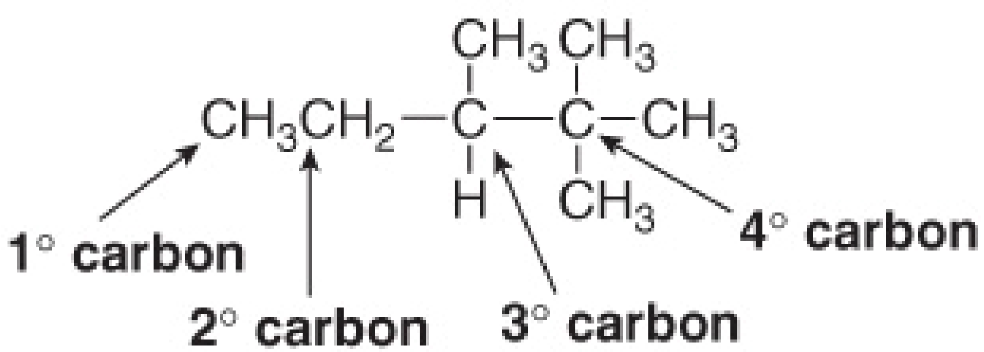


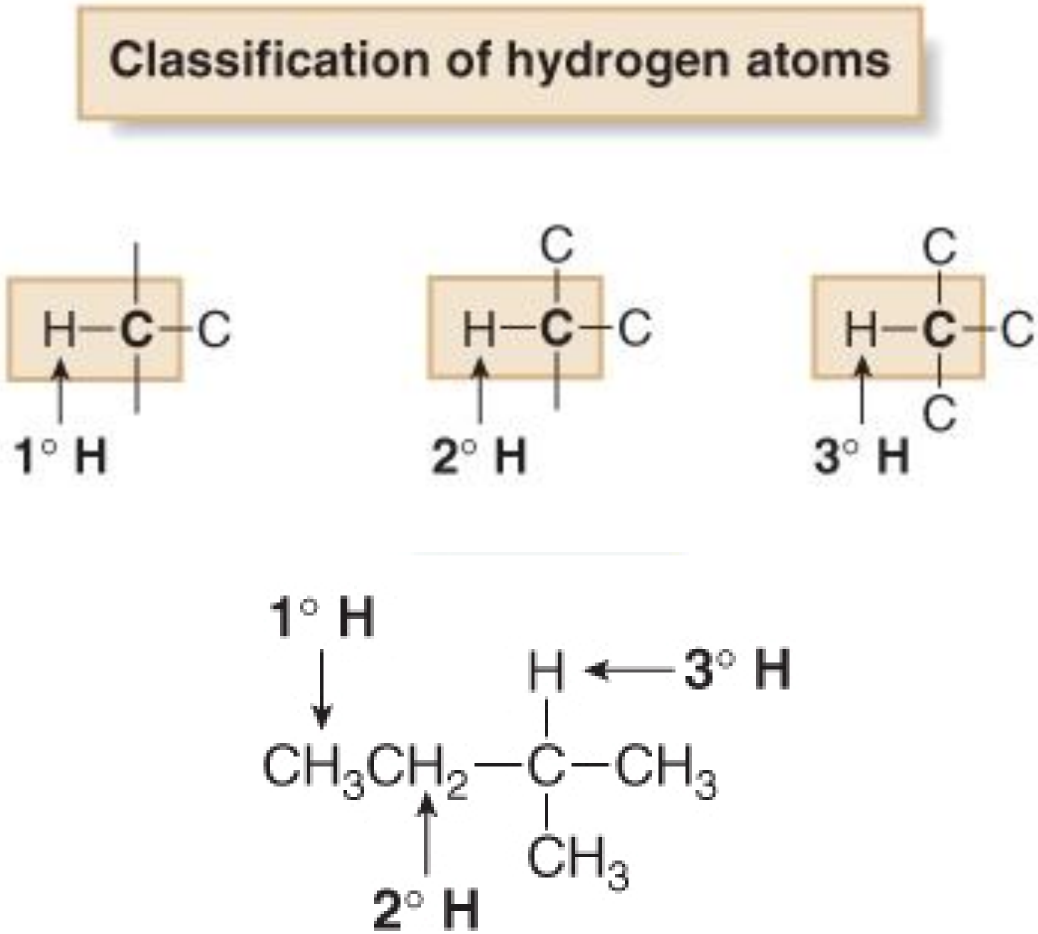
2,2-dimethylpropane

Classification of carbon atoms



- A *primary carbon* (1° carbon) is bonded to *one* other C atom.
- A *secondary carbon* (2° carbon) is bonded to *two* other C atoms.
- A *tertiary carbon* (3° carbon) is bonded to *three* other C atoms.
- A *quaternary carbon* (4° carbon) is bonded to *four* other C atoms.

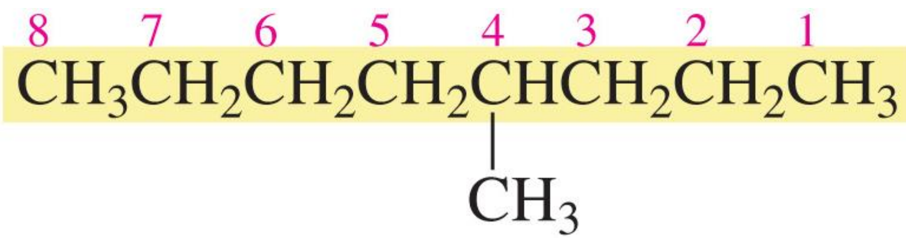




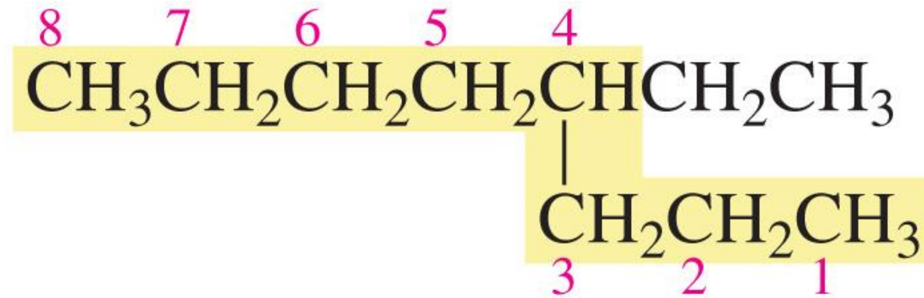
- Hydrogen atoms are classified as primary (1^{o}), secondary (2^{o}), or tertiary (3^{o}) depending on the type of carbon atom to which they are bonded.

Nomenclature of Alkanes

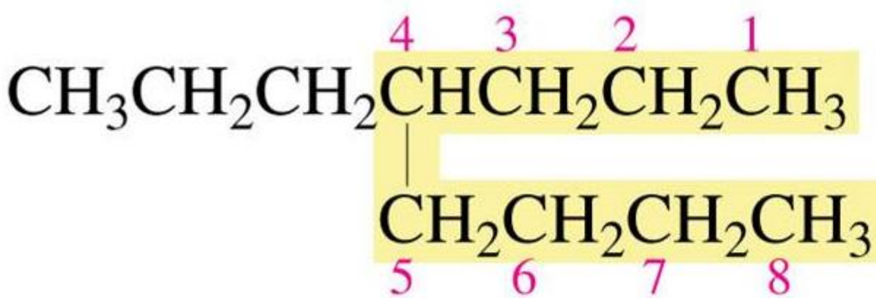
1. Determine the number of carbons in the longest continuous chain



4-methyloctane

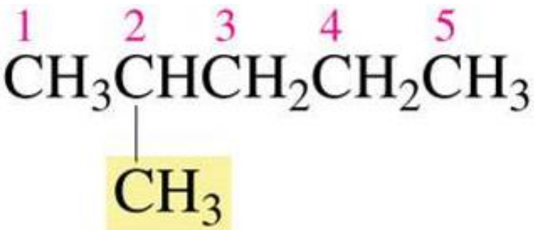


4-ethyloctane

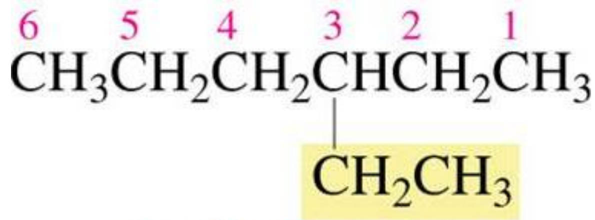


4-propyloctane

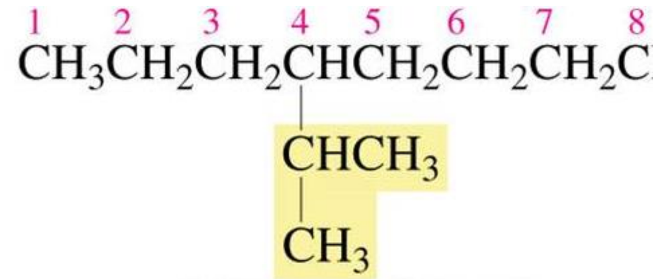
2. Number the chain so that the substituent gets the lowest number



2-methylpentane

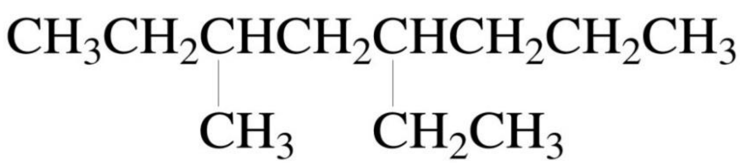


3-ethylhexane



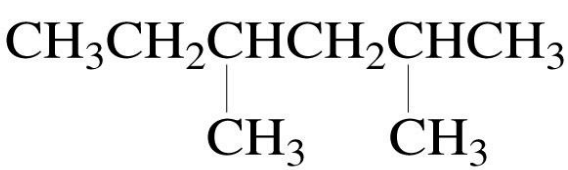
4-isopropyloctane

3. Number the substituents to yield the lowest possible number in the number of the compound

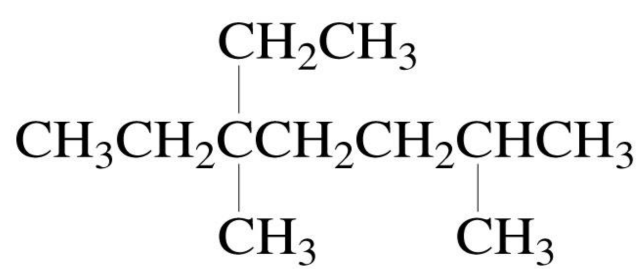


5-ethyl-3-methyloctane
not
4-ethyl-6-methyloctane
because 3 < 4

substituents are listed in alphabetical order

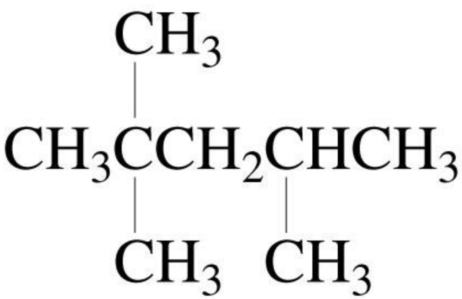


2,4-dimethylhexane

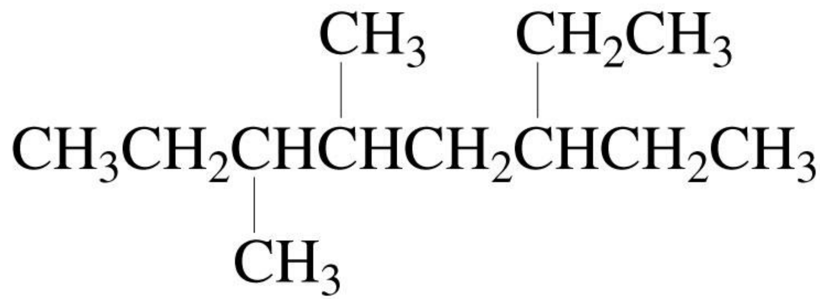


5-ethyl-2,5-dimethylheptane

4. Assign the lowest possible numbers to all of the substituents

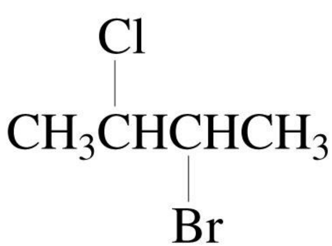


2,2,4-trimethylpentane
not
2,4,4-trimethylpentane
because 2 < 4

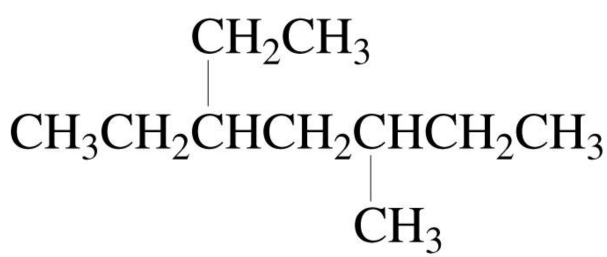


6-ethyl-3,4-dimethyloctane
not
3-ethyl-5,6-dimethyloctane
because 4 < 5

5. If the same substituent numbers are obtained in both directions, the first group cited receives the lower number

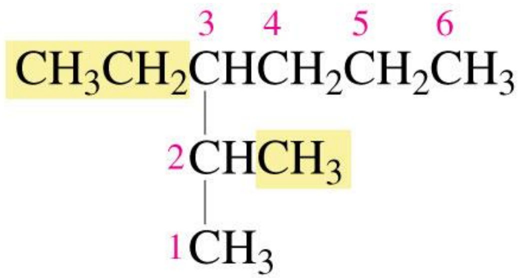


2-bromo-3-chlorobutane
not
3-bromo-2-chlorobutane

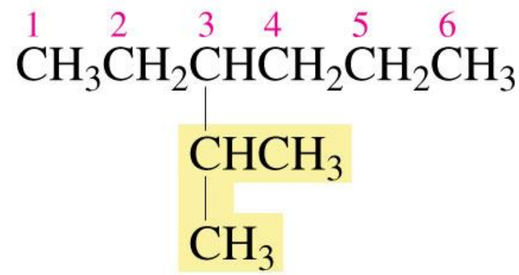


3-ethyl-5-methylheptane
not
5-ethyl-3-methylheptane

6. If a compound has two or more chains of the same length, the parent hydrocarbon is the chain with the greatest number of substituents



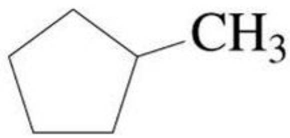
3-ethyl-2-methylhexane (two substituents)



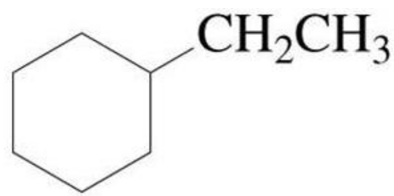
not
3-isopropylhexane (one substituent)

Nomenclature of Cycloalkanes

1. No number is needed for a single substituent on a ring



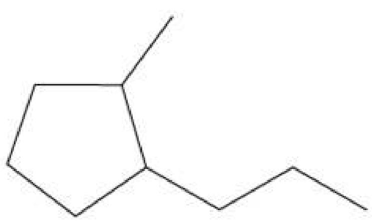
methylcyclopentane



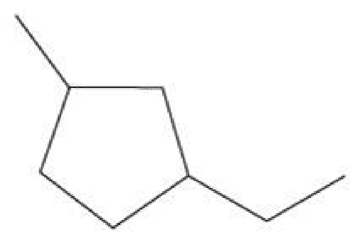
ethylcyclohexane

1-cyclobutylpentane

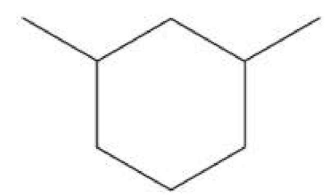
2. Name the two substituents in alphabetical order



1-methyl-2-propylcyclopentane

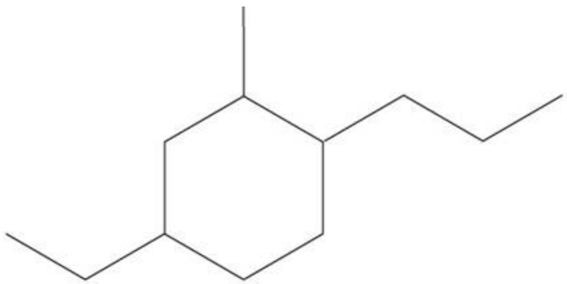


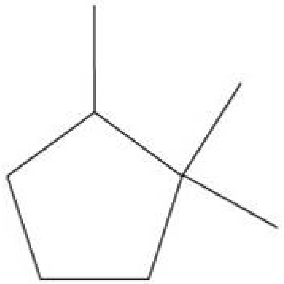
1-ethyl-3-methylcyclopentane



1,3-dimethylcyclohexane

3. If there are more than two substituents, they are cited in alphabetical order





4-ethyl-2-methyl-1-propylcyclohexane

not

1-ethyl-3-methyl-4-propylcyclohexane

because 2 < 3

not

5-ethyl-1-methyl-2-propylcyclohexane

because 4 < 5

1,1,2-trimethylcyclopentane

not

1,2,2-trimethylcyclopentane

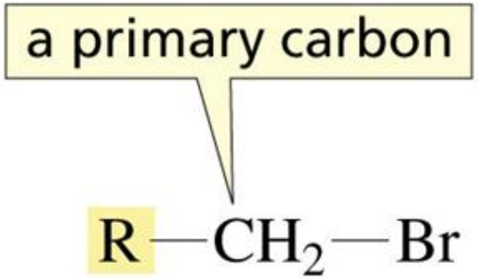
because 1 < 2

not

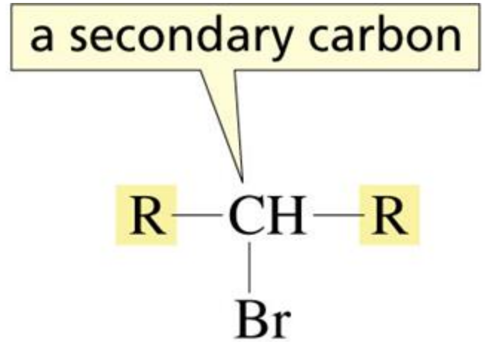
1,1,5-trimethylcyclopentane

because 2 < 5

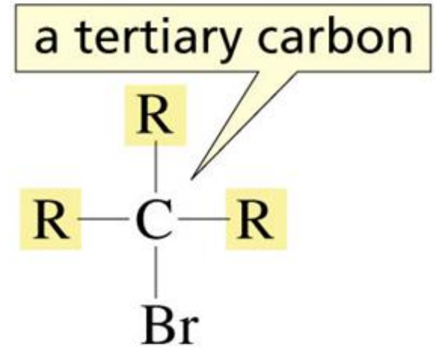
Nomenclature of Alkyl Halides



a primary alkyl halide

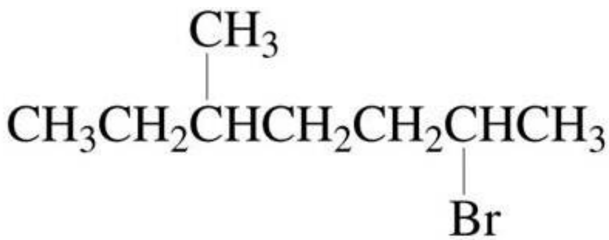


a secondary alkyl halide



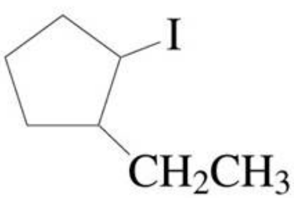
a tertiary alkyl halide

In the IUPAC system, alkyl halides are named as substituted alkanes

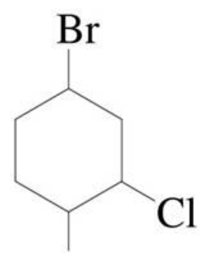


2-bromo-5-methylheptane

1-chloro-6,6-dimethylheptane



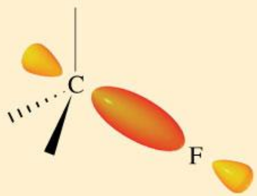
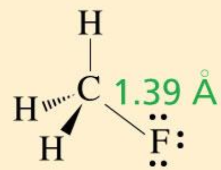
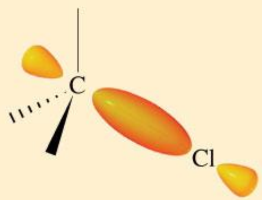
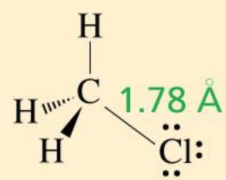
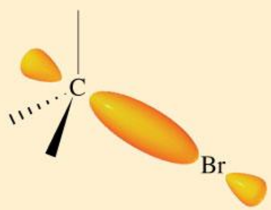
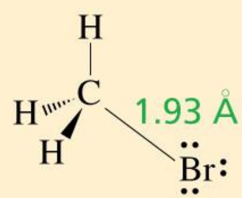
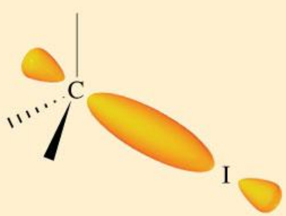
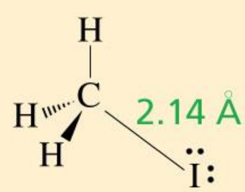
1-ethyl-2-iodocyclopentane



4-bromo-2-chloro-1-methylcyclohexane

Structures of **Alkyl Halides**

Table 2.4 Carbon–Halogen bond lengths and bond strengths

	Orbital interactions	Bond lengths	Bond strength (kcal/mol)	(kJ/mol)
$H_3C—F$			108	451
$H_3C—Cl$			84	350
$H_3C—Br$			70	294
$H_3C—I$			57	239

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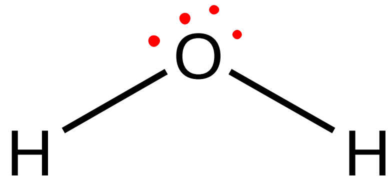
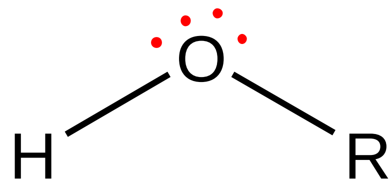
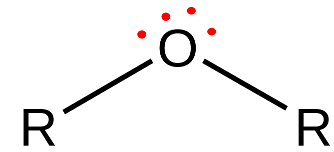
Structure of alcohol and amine



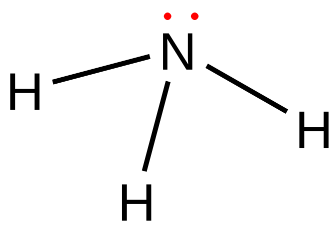
Water



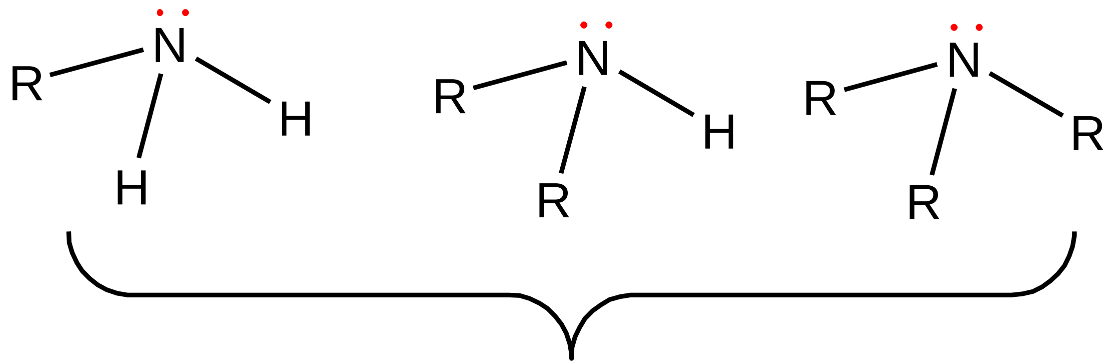
Alcohol



Ether



Amonia



Amines

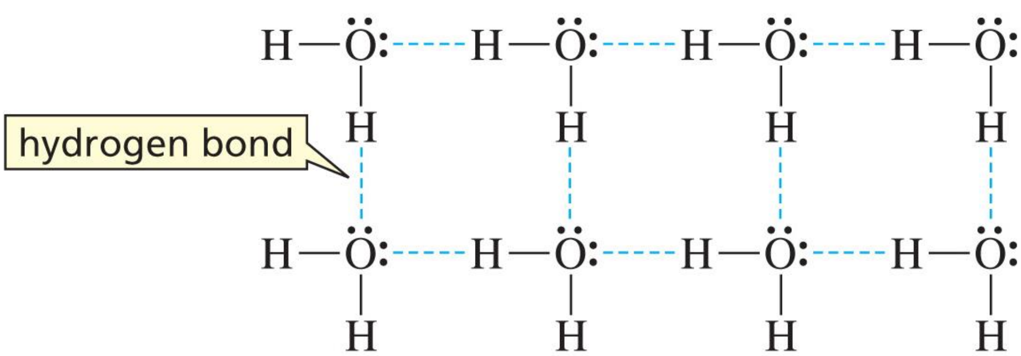
Physical Properties of Alkanes

Hydrogen bonding, dipole-dipole interaction and **Van der Waals** attractions are examples of <u>Nonbonding intermolecular interactions.</u>

1. Solubility in water: Alkanes are insoluble in H_2O why?

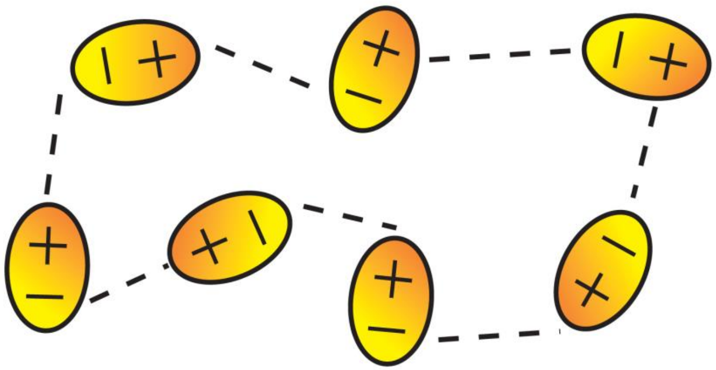
The small electronegativity difference between C and H is ignored.

Alkanes are non-Polar compound



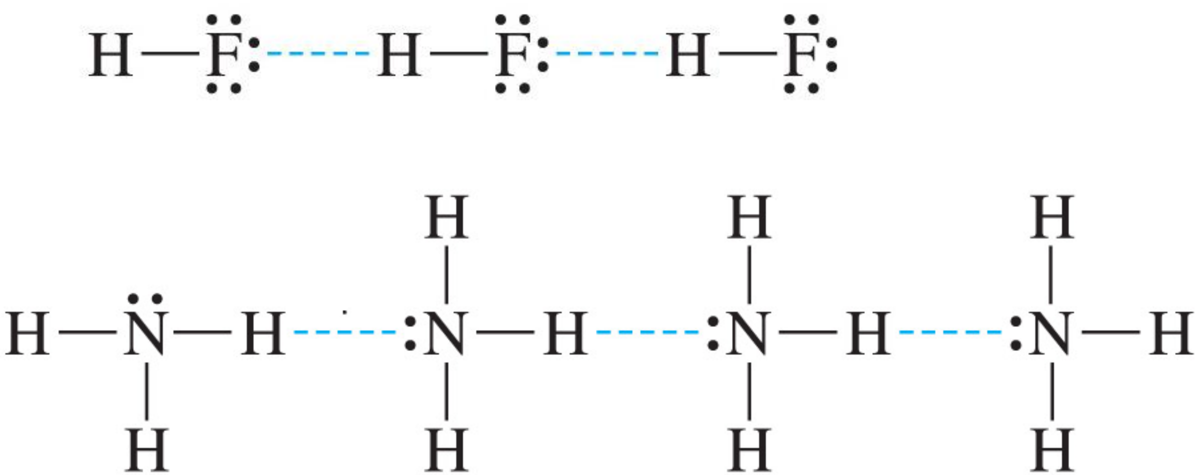
Likes dissolves likes

Solubility in water

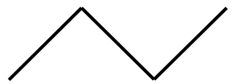


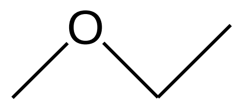
HCl, alkyl halides, ether

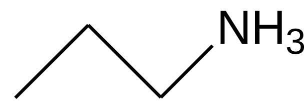
Dipole-dipole interaction

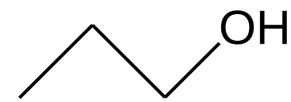


Hydrogen bond









boiling points (°C) -0.5 10.8 47.8 97.4

2. Boiling points:

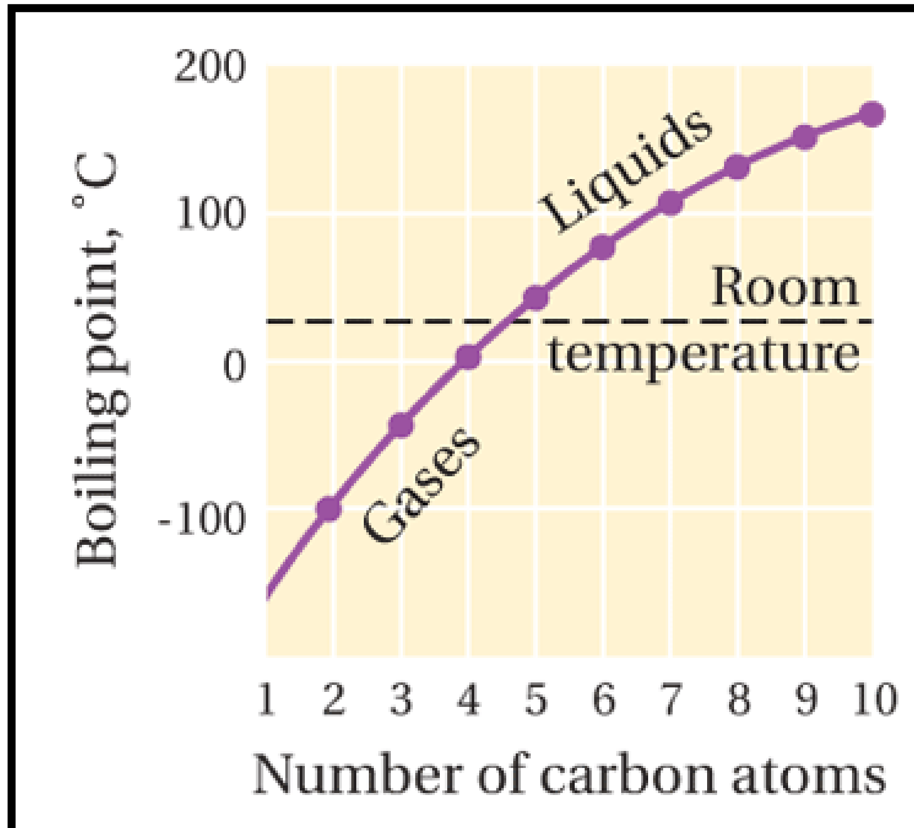
<u>General rule:</u> Straight-chain alkanes

*The boiling points of alkanes rise as the **<u>chain length</u>** increases*

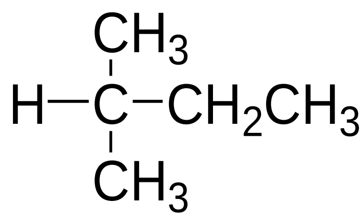
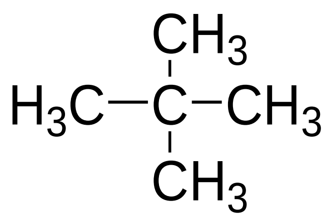
Q: which has the highest boiling point?

Butane: $CH_3CH_2CH_2CH_2$ ***Or***

pentane: $CH_3CH_2CH_2CH_2CH_2$



Q: Despite having the same molecular weight, the boiling point decreases as you go down from *n*-pentane to 2,2-dimethylpropane. Explain?

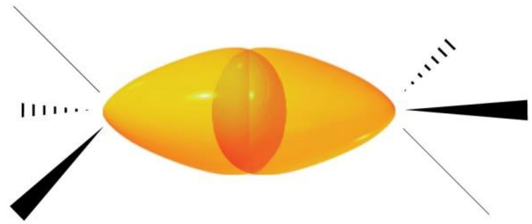
IUPAC name	Formula	Boiling point (B.P)
Pentane	$CH_3(CH_2)_3CH_3$	36
2-methylbutane		28
2,2-dimethylpropane		10

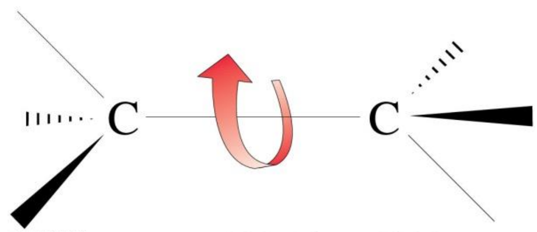
Conclusion: As branching increases, boiling point decreases

Table 2.1 Nomenclature and Physical Properties of Straight-Chain Alkanes						
Number of carbons	**Molecular formula**	**Name**	**Condensed structure**	**Boiling point (°C)**	**Melting point (°C)**	**Density[a] (g/mL)**
1	CH_4	methane	CH_4	−167.7	−182.5	
2	C_2H_6	ethane	CH_3CH_3	−88.6	−183.3	
3	C_3H_8	propane	$CH_3CH_2CH_3$	−42.1	−187.7	
4	C_4H_{10}	butane	$CH_3CH_2CH_2CH_3$	−0.5	−138.3	
5	C_5H_{12}	pentane	$CH_3(CH_2)_3CH_3$	36.1	−129.8	0.5572
6	C_6H_{14}	hexane	$CH_3(CH_2)_4CH_3$	68.7	−95.3	0.6603
7	C_7H_{16}	heptane	$CH_3(CH_2)_5CH_3$	98.4	−90.6	0.6837
8	C_8H_{18}	octane	$CH_3(CH_2)_6CH_3$	125.7	−56.8	0.7026
9	C_9H_{20}	nonane	$CH_3(CH_2)_7CH_3$	150.8	−53.5	0.7177
10	$C_{10}H_{22}$	decane	$CH_3(CH_2)_8CH_3$	174.0	−29.7	0.7299
11	$C_{11}H_{24}$	undecane	$CH_3(CH_2)_9CH_3$	195.8	−25.6	0.7402
12	$C_{12}H_{26}$	dodecane	$CH_3(CH_2)_{10}CH_3$	216.3	−9.6	0.7487
13	$C_{13}H_{28}$	tridecane	$CH_3(CH_2)_{11}CH_3$	235.4	−5.5	0.7546
⋮	⋮	⋮	⋮	⋮	⋮	⋮
20	$C_{20}H_{42}$	eicosane	$CH_3(CH_2)_{18}CH_3$	343.0	36.8	0.7886
21	$C_{21}H_{44}$	heneicosane	$CH_3(CH_2)_{19}CH_3$	356.5	40.5	0.7917
⋮	⋮	⋮	⋮	⋮	⋮	⋮
30	$C_{30}H_{62}$	triacontane	$CH_3(CH_2)_{28}CH_3$	449.7	65.8	0.8097

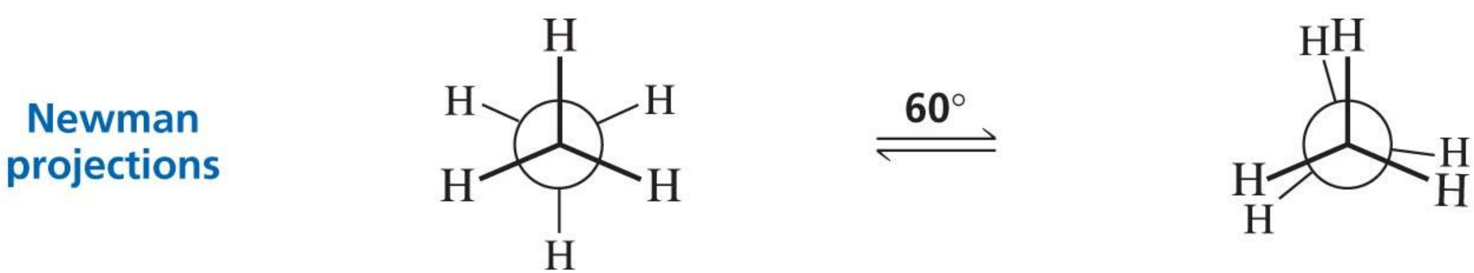
[a]Density is temperature dependent. The densities given are those determined at 20 °C ($d^{20°}$).

Conformations of Alkanes: Rotation about Carbon–Carbon Bonds





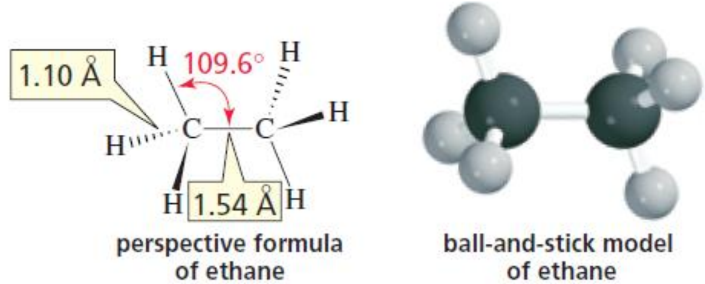
$H_3C—CH_3$
ethane

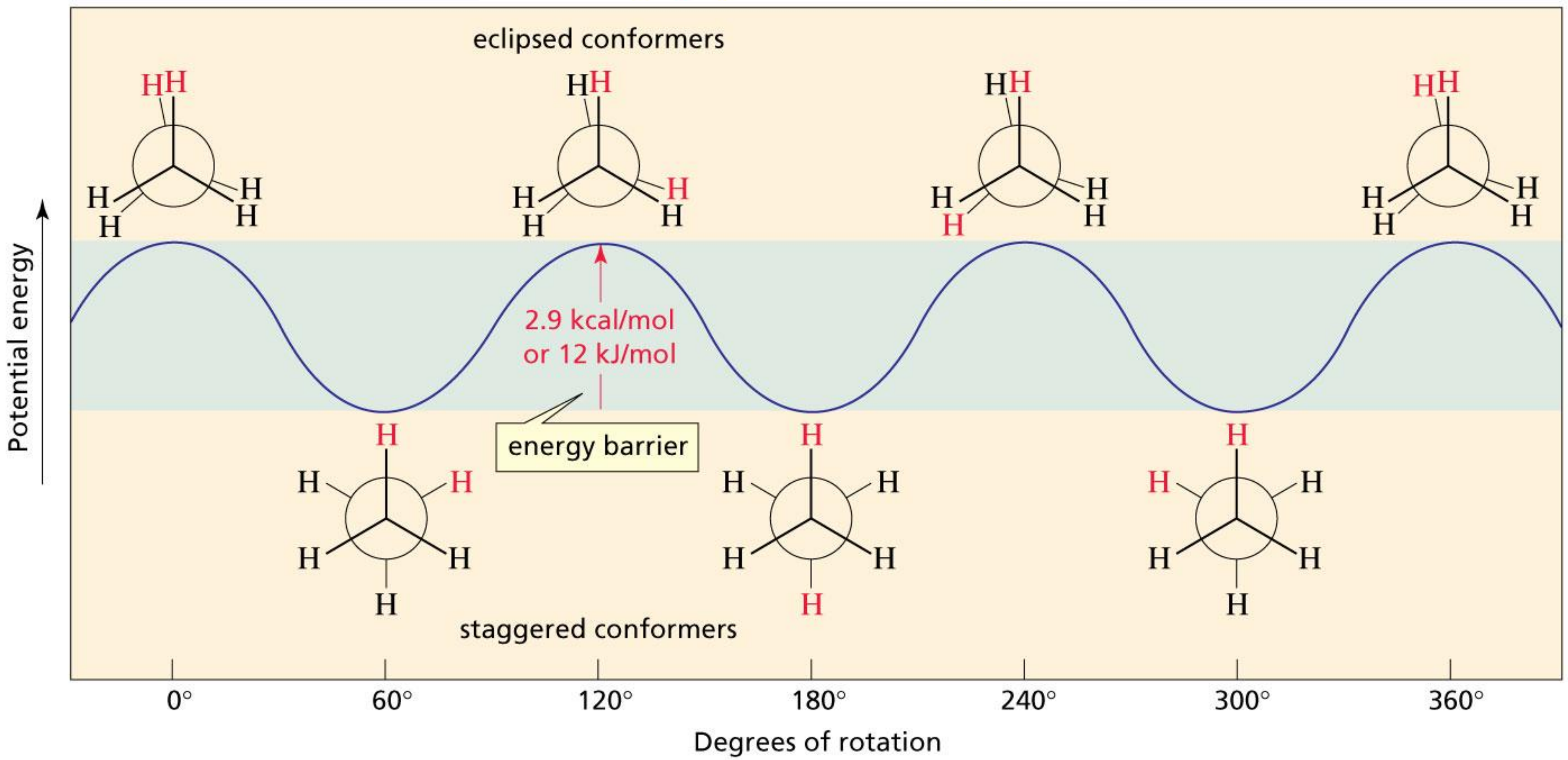


staggered conformer from rotation about the C — C bond in ethane

eclipsed conformer from rotation about the C — C bond in ethane

Different Conformations of **Ethane**





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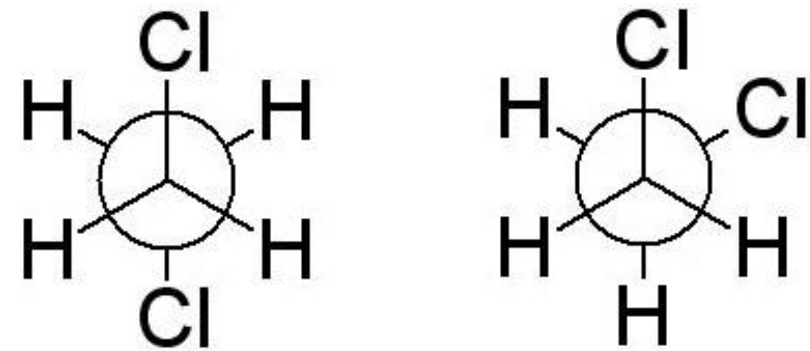
Q: What is the relationship between the Newman representations shown?



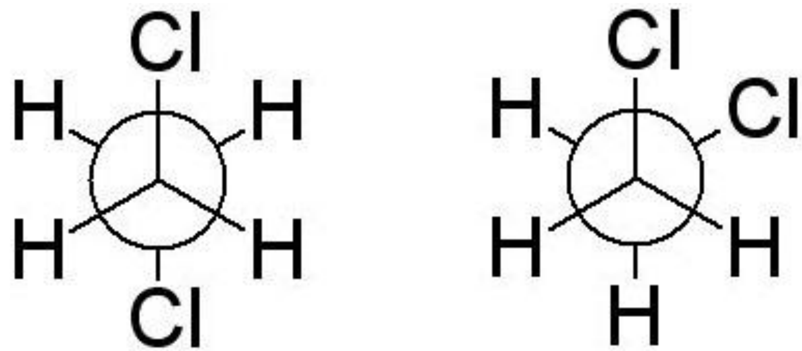
A) identical

B) constitutional isomers

C) Conformational isomer

D) stereoisomers

Q: What is the relationship between the Newman representations shown?



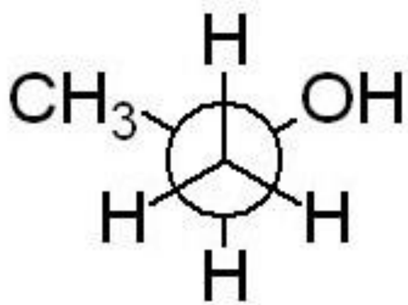
A) identical

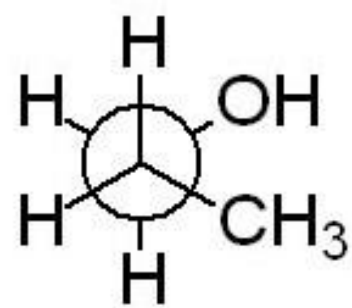
B) constitutional isomers

C) Conformational isomer

D) stereoisomers

Q: What is the relationship between the Newman representations shown?





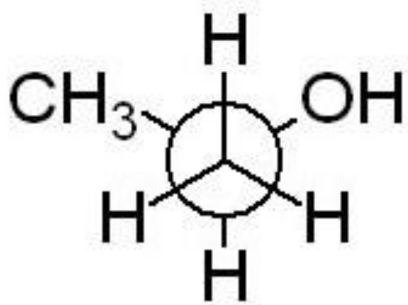
A) Identical

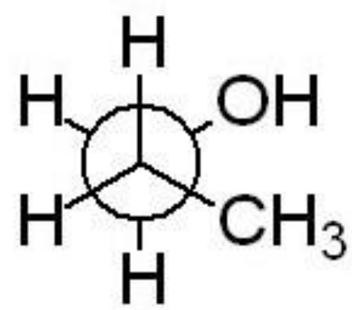
B) constitutional isomers

C) Conformational isomer

D) stereoisomers

Q: What is the relationship between the Newman representations shown?





A) Identical

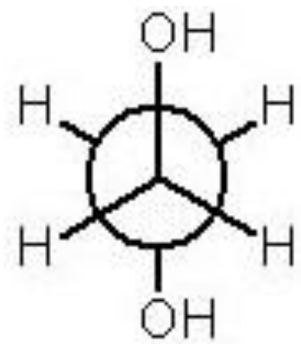
B) constitutional isomers

C) Conformational isomer

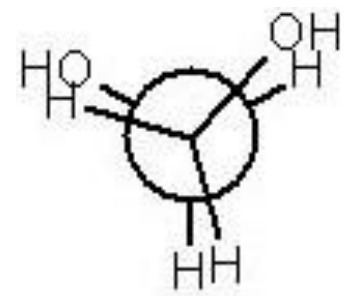
D) stereoisomers

Q: Select the least stable conformation of ethylene glycol.

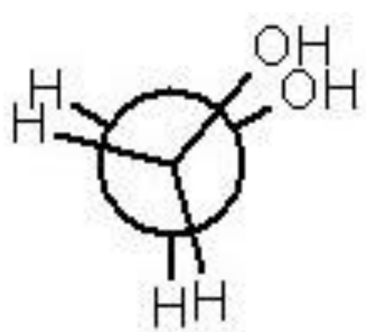
A)



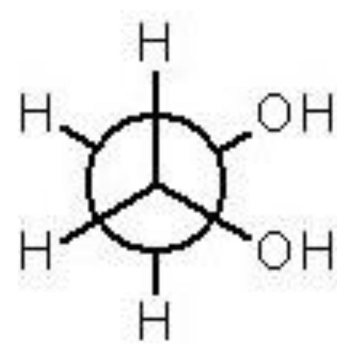
B)



C)

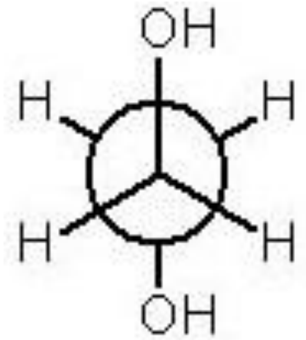


D)

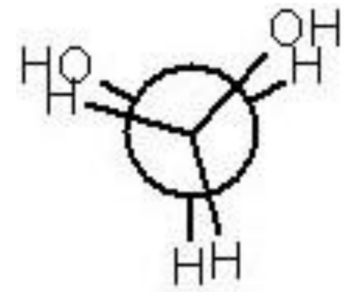


Q: Select the <u>least</u> stable conformation of ethylene glycol.

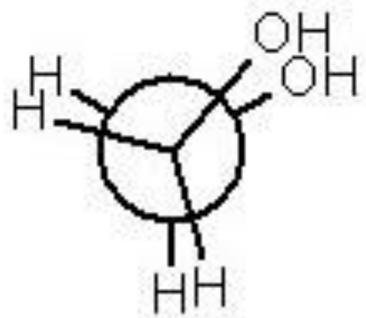
A)



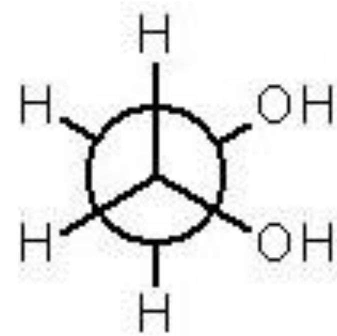
B)



C)

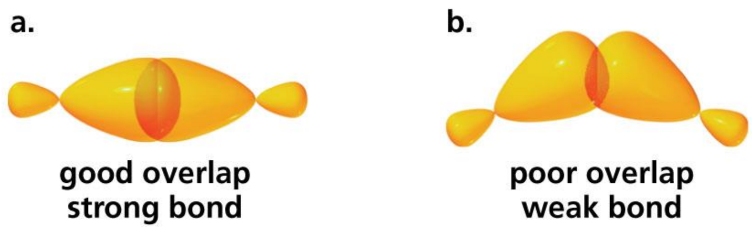


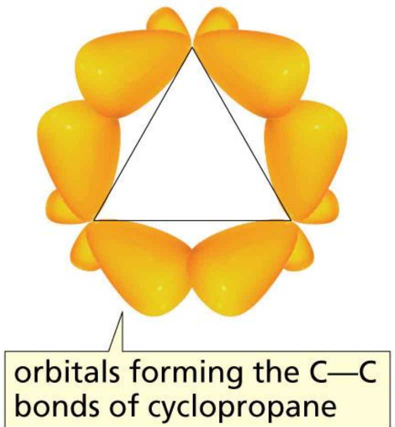
D)

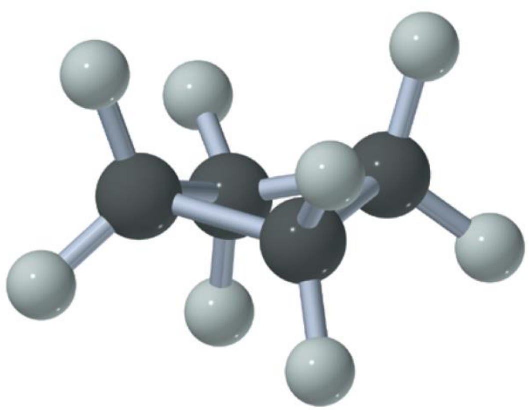


Cycloalkanes: **Ring Strain**

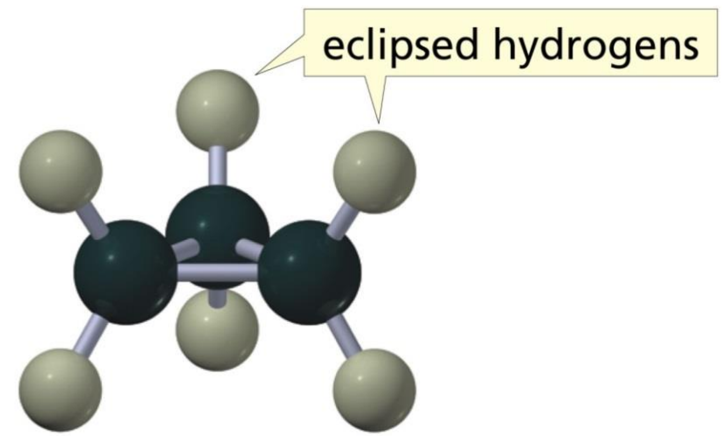
- Angle strain results when bond angles deviate from the ideal 109.5° bond angle





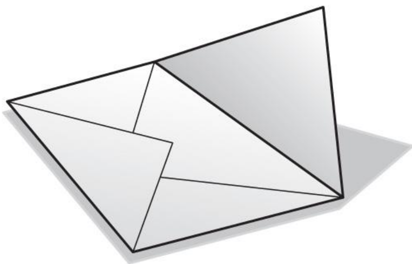


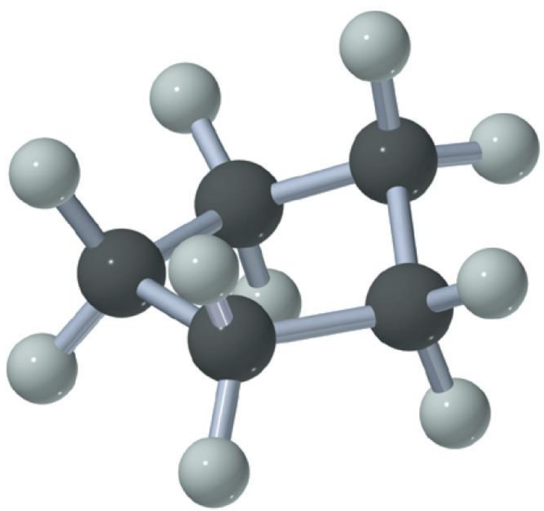
cyclobutane



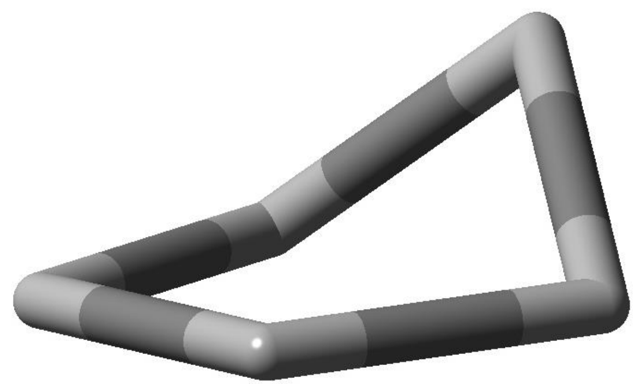
cyclopropane

Cycloalkanes: **Ring Strain**

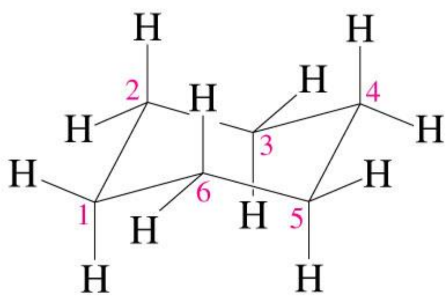




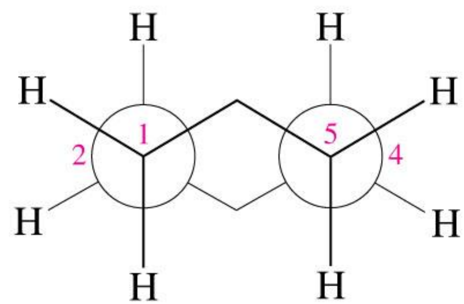
cyclopentane



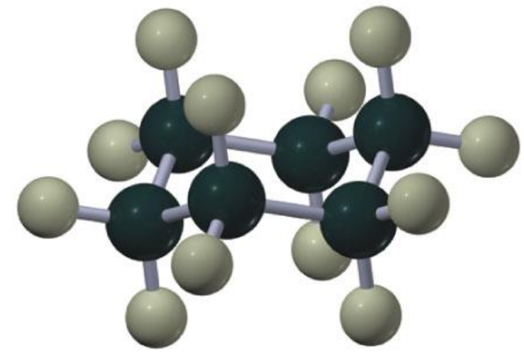
The chair conformation of **cyclohexane** is free of strain



chair conformer of cyclohexane

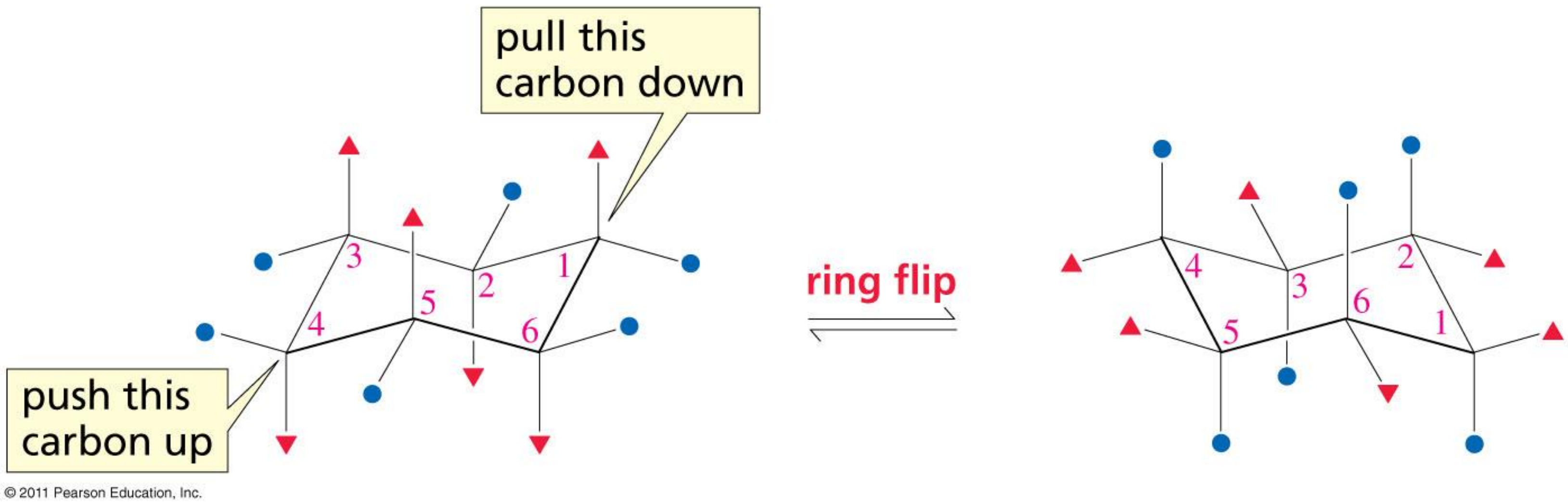


Newman projection of the chair conformer

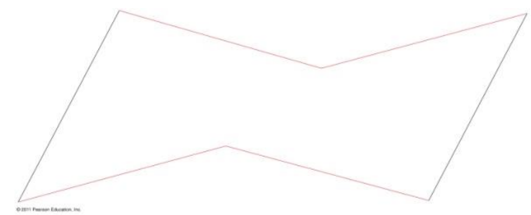


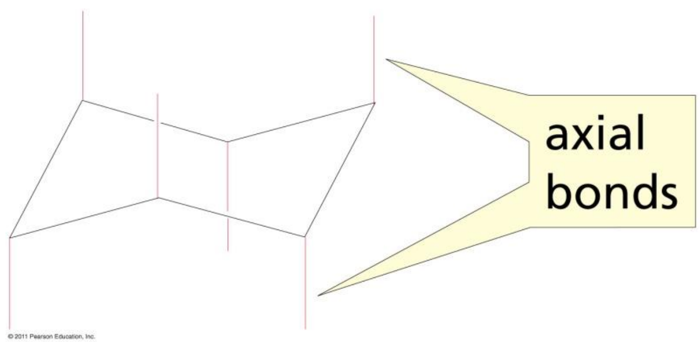
ball-and-stick model of the chair conformer

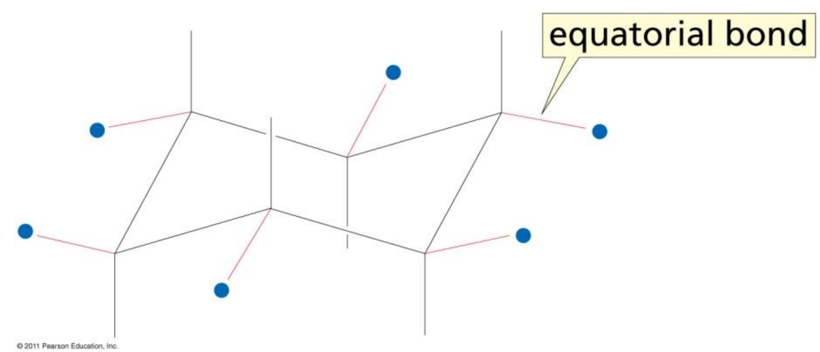
Ring Flipping in **Cyclohexane**

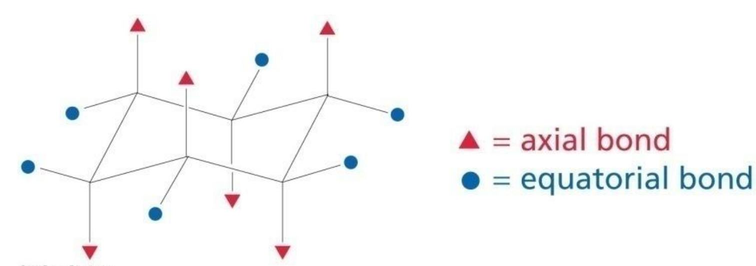


Drawing **Cyclohexane**

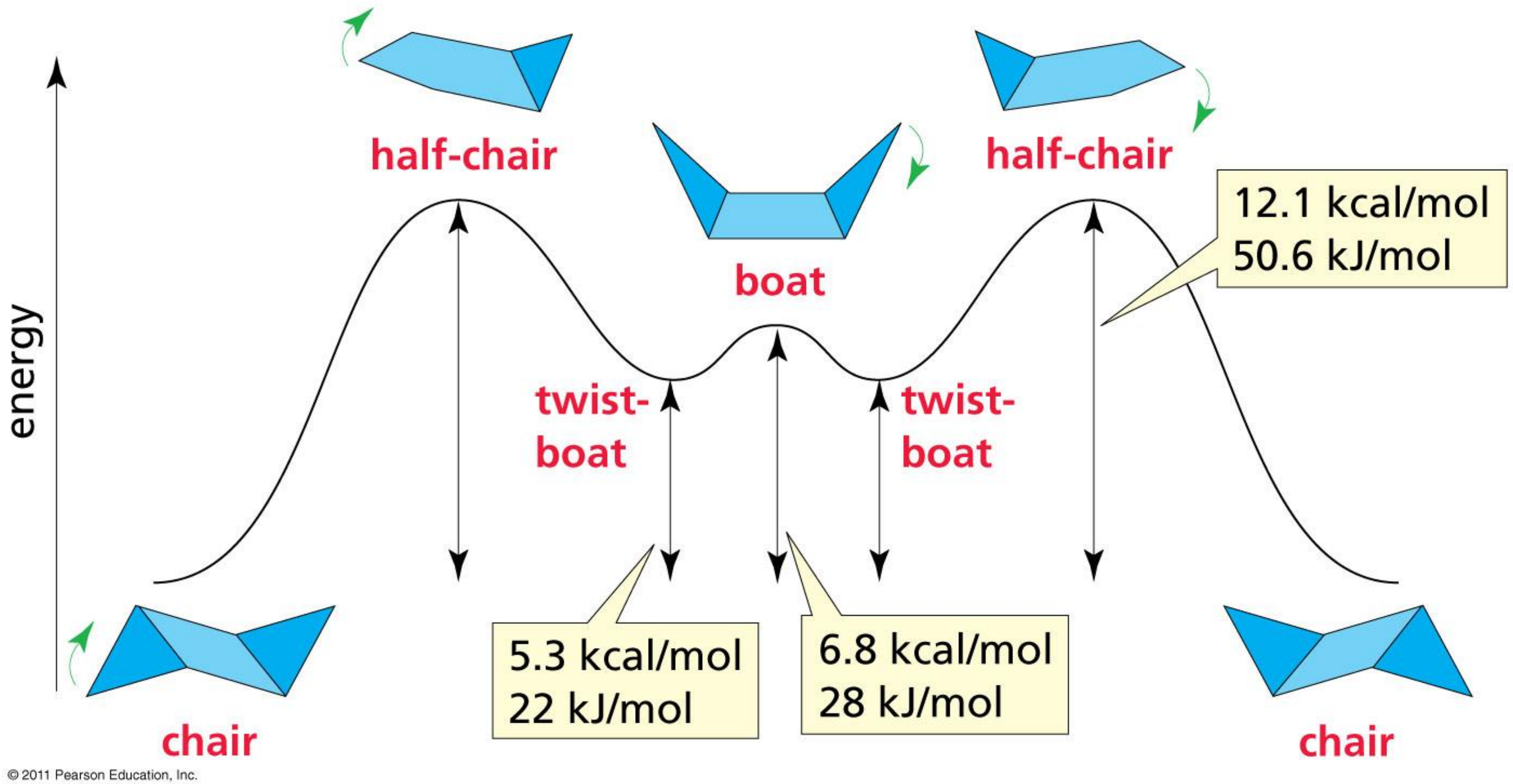






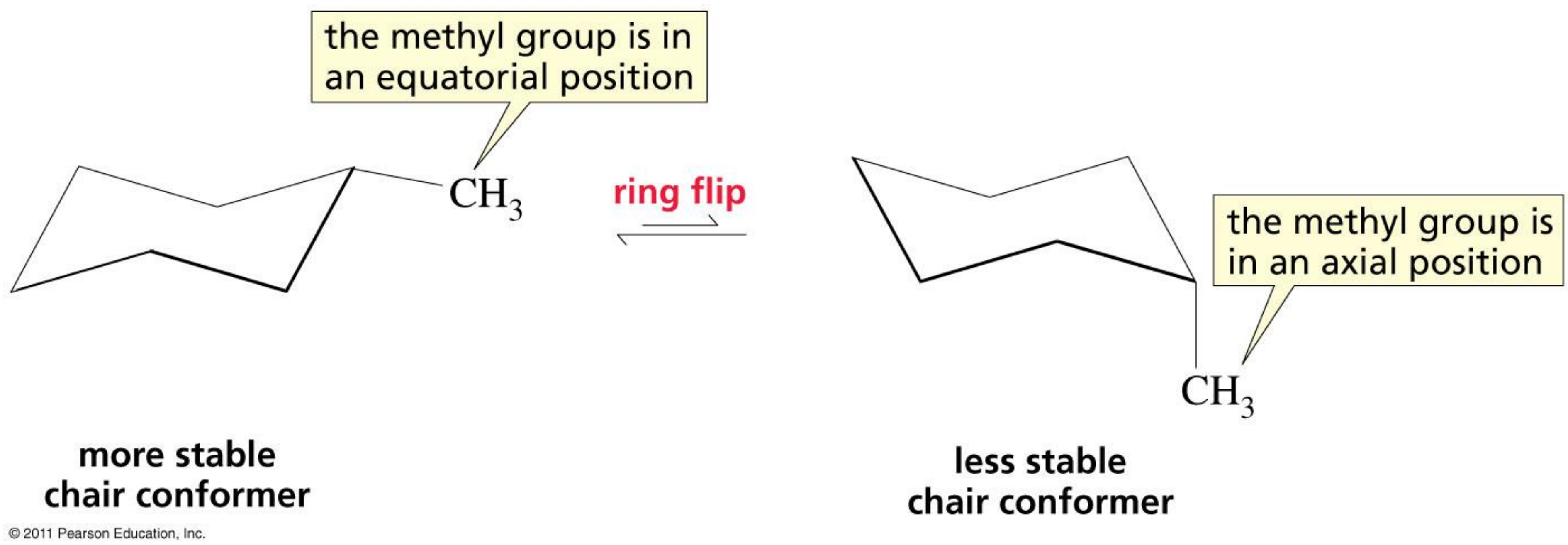


The Conformations of **Cyclohexane** and Their Energies

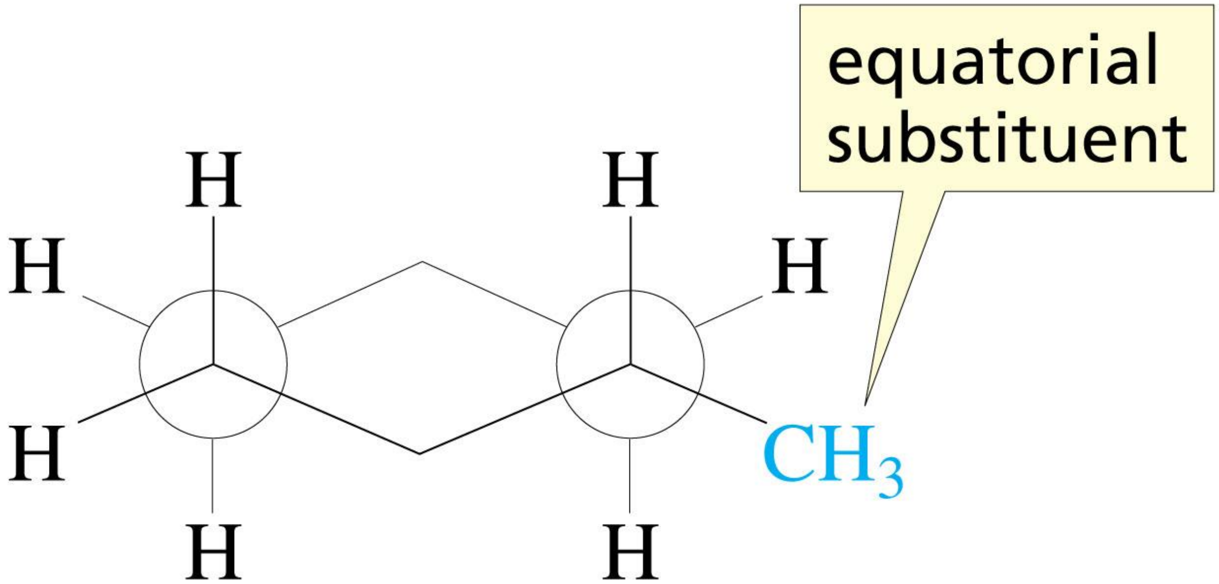


http://www.youtube.com/watch?v=bPLREpfZ63I

Conformations of Monosubstituted **Cyclohexanes**

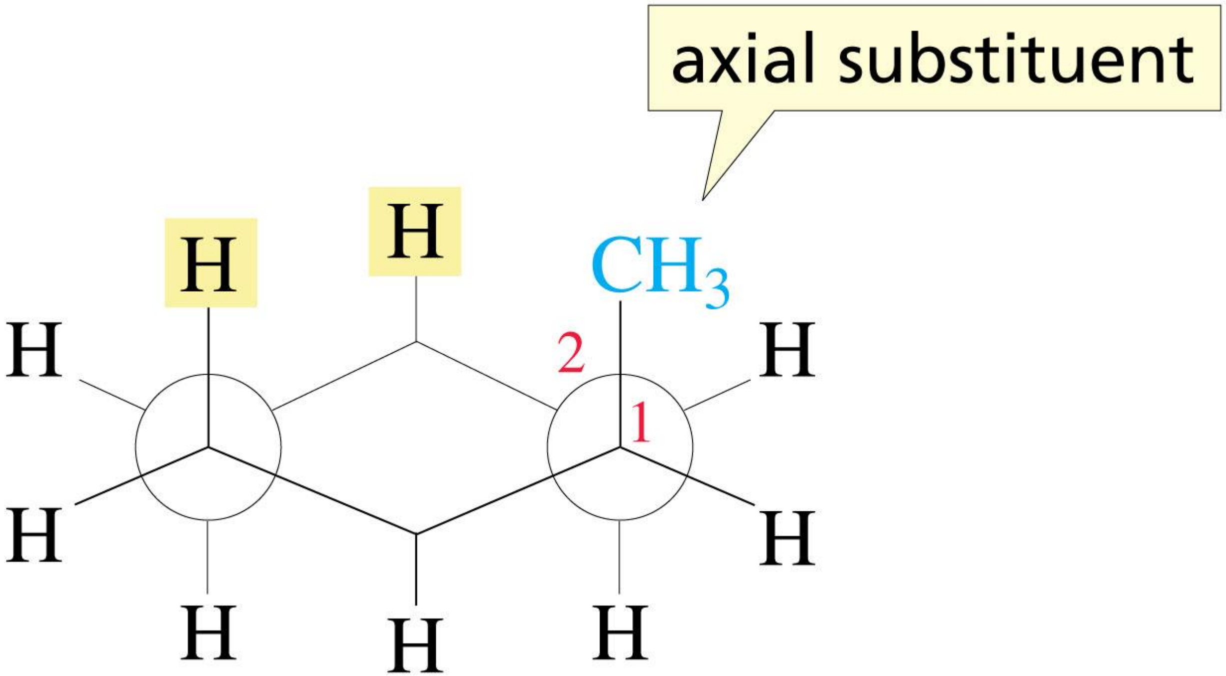


a.



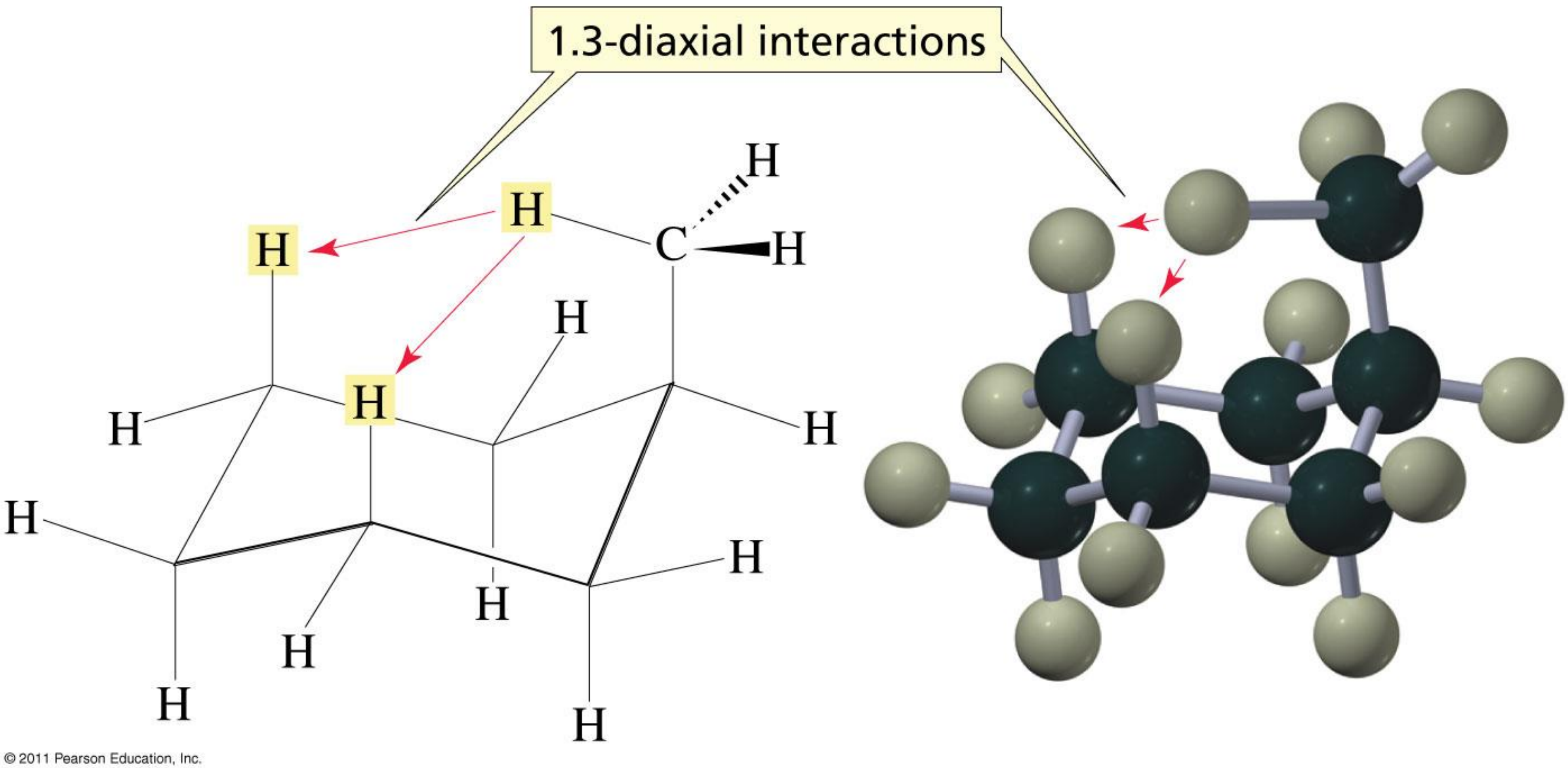
equatorial substituent
H
H
H
H
H
H
H
H
CH3

b.



axial substituent
H
H
CH3
H
2
H
1
H
H
H
H
H
H

Steric Strain of **1,3-Diaxial** Interaction in **Methylcyclohexane**



The larger the substituent on a cyclohexane ring, the more the equatorial substituted conformer will be favored

Table 2.10 Equilibrium Constants for Several Monosubstituted Cyclohexanes at 25 °C

Substituent	**Axial** $\xrightleftharpoons{K_{eq}}$ **Equatorial**	**Substituent**	**Axial** $\xrightleftharpoons{K_{eq}}$ **Equatorial**
H	1	CN	1.4
CH_3	18	F	1.5
CH_3CH_2	21	Cl	2.4
$CH_3\overset{\overset{CH_3}{\vert}}{C}H$	35	Br	2.2
$CH_3\underset{\underset{CH_3}{\vert}}{\overset{\overset{CH_3}{\vert}}{C}}$	4800	I	2.2
		HO	5.4

K_{eq} = [equatorial conformer]/[axial conformer]

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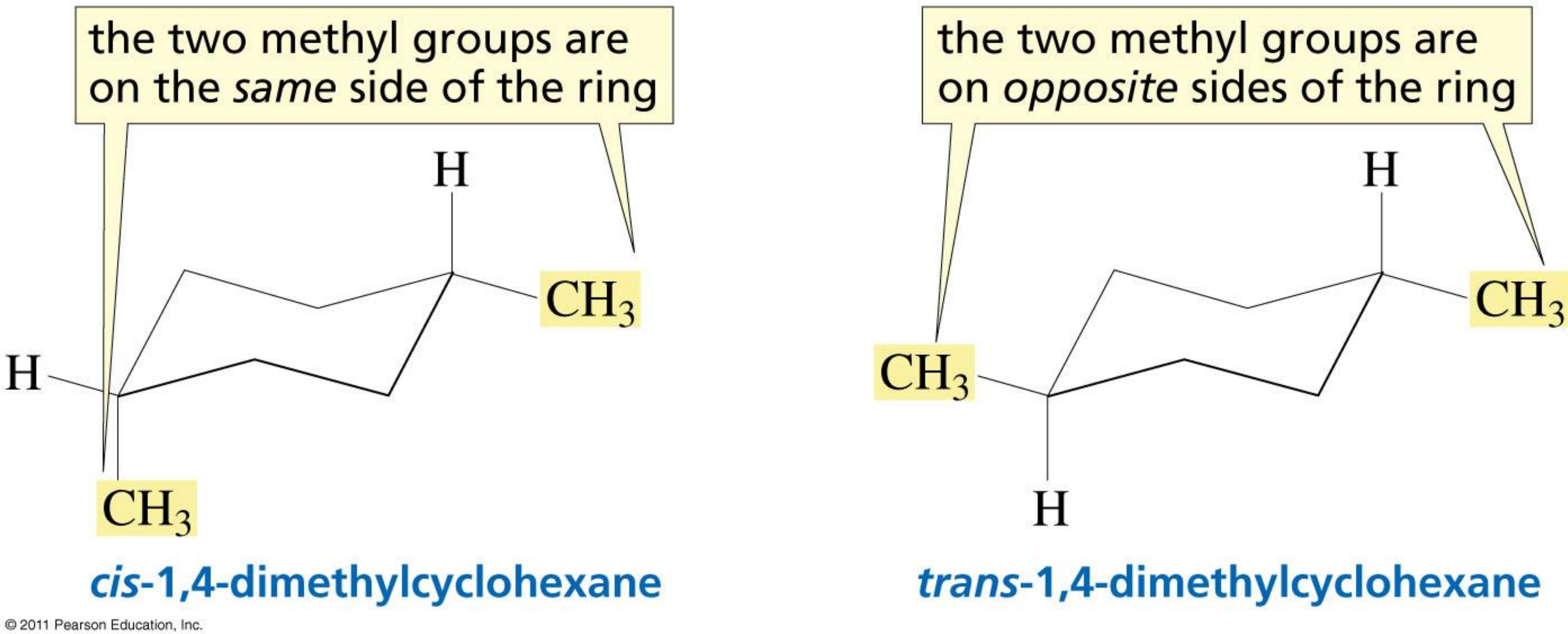
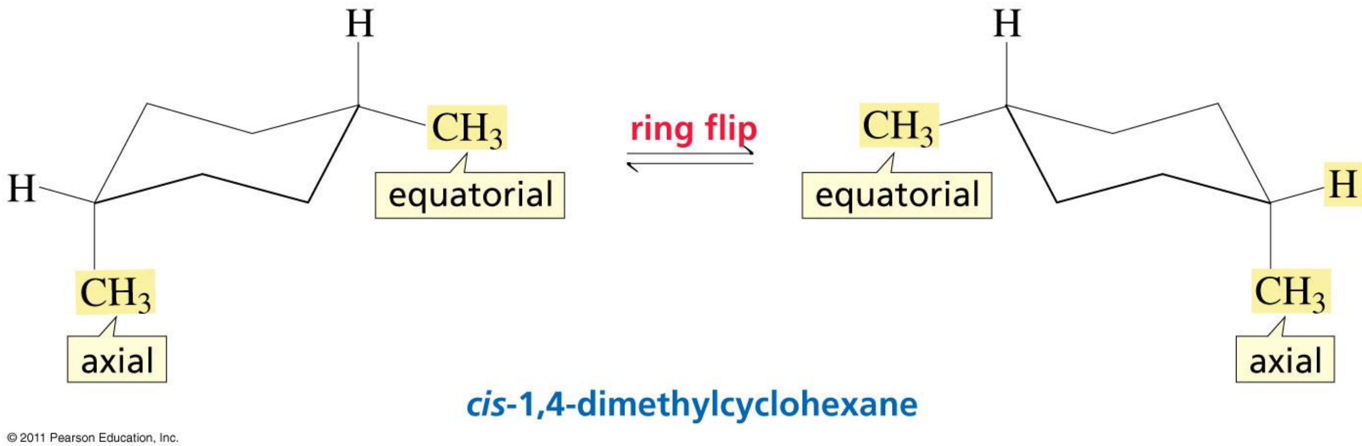
The Chair Conformers of ***cis***- or ***trans***-1,4-Dimethylcyclohexane




H
CH_3
equatorial
ring flip
CH_3
equatorial
H
H
H
CH_3
axial
CH_3
axial
cis-1,4-dimethylcyclohexane
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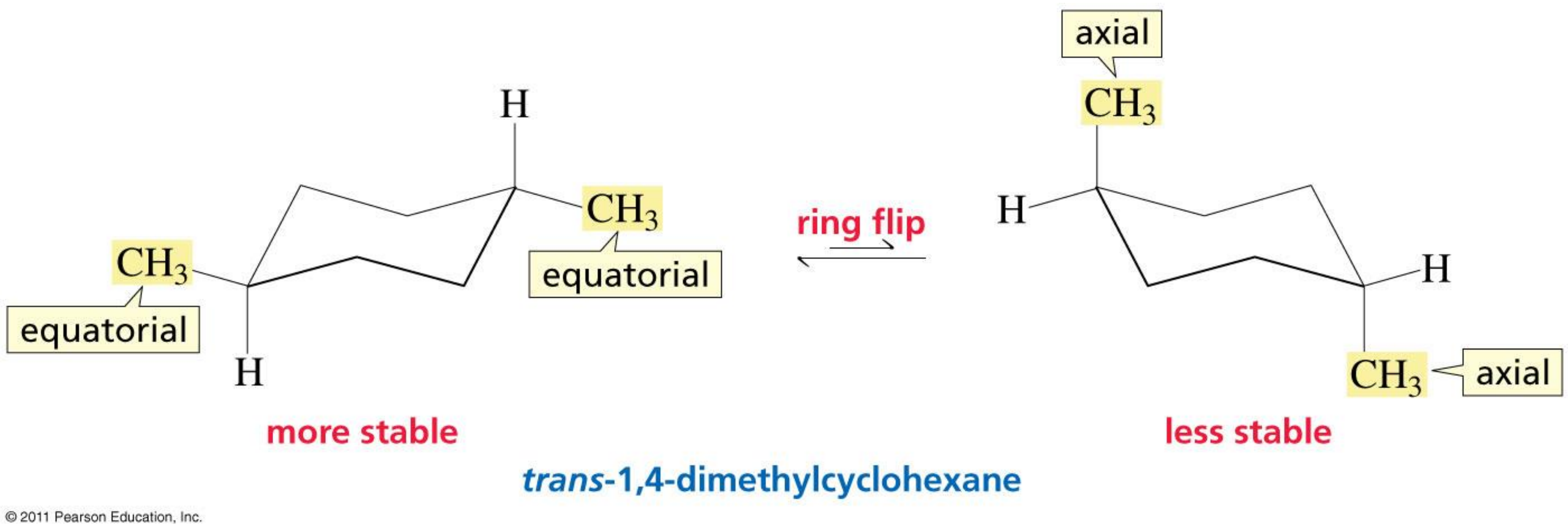
axial
CH_3
H
CH_3
equatorial
ring flip
H
CH_3
equatorial
H
H
CH_3
axial
more stable
less stable
trans-1,4-dimethylcyclohexane
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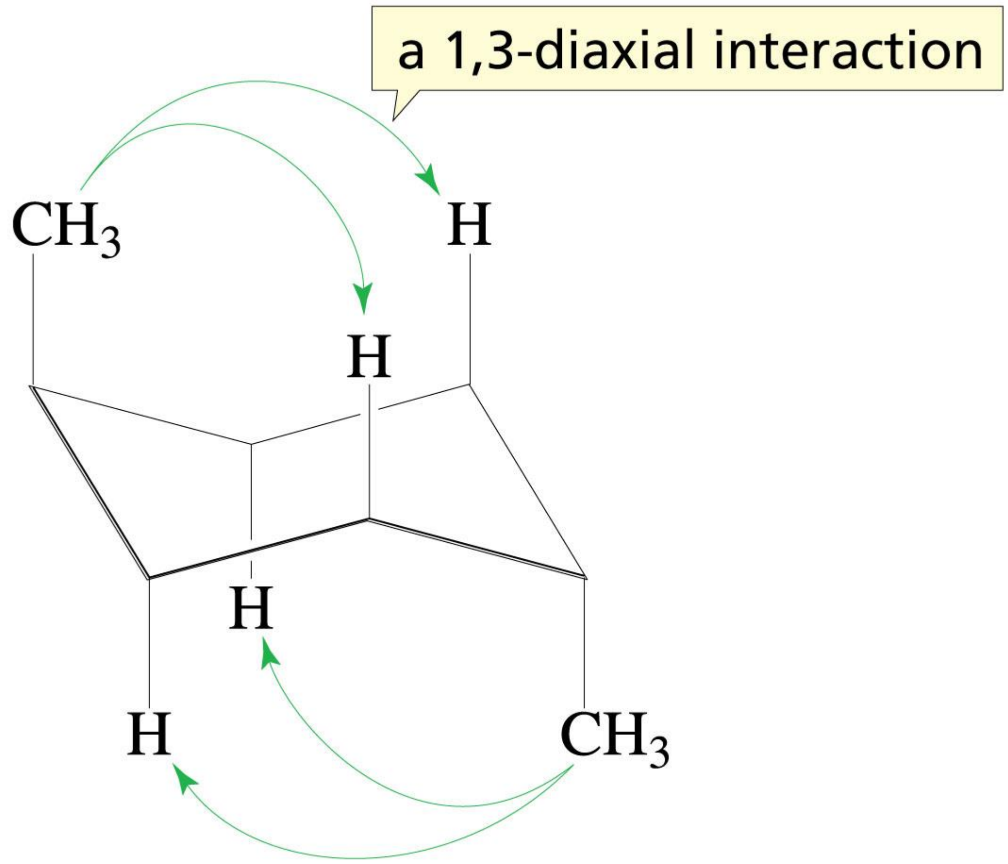
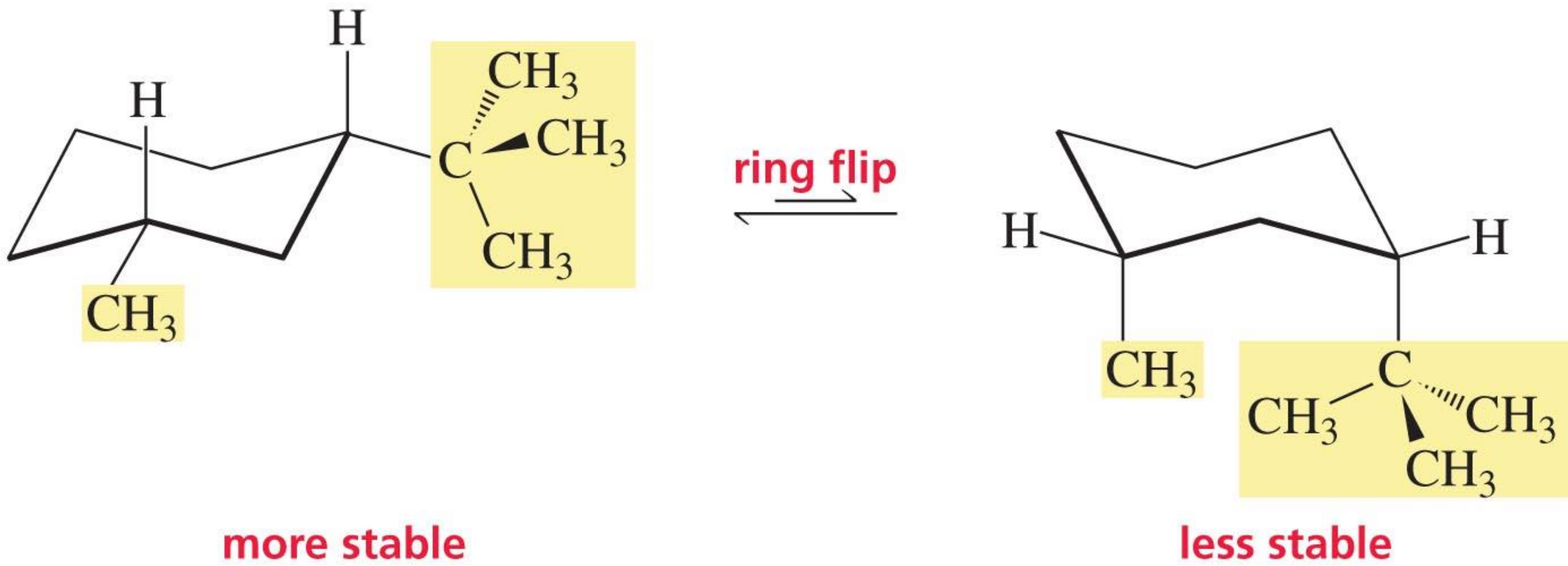
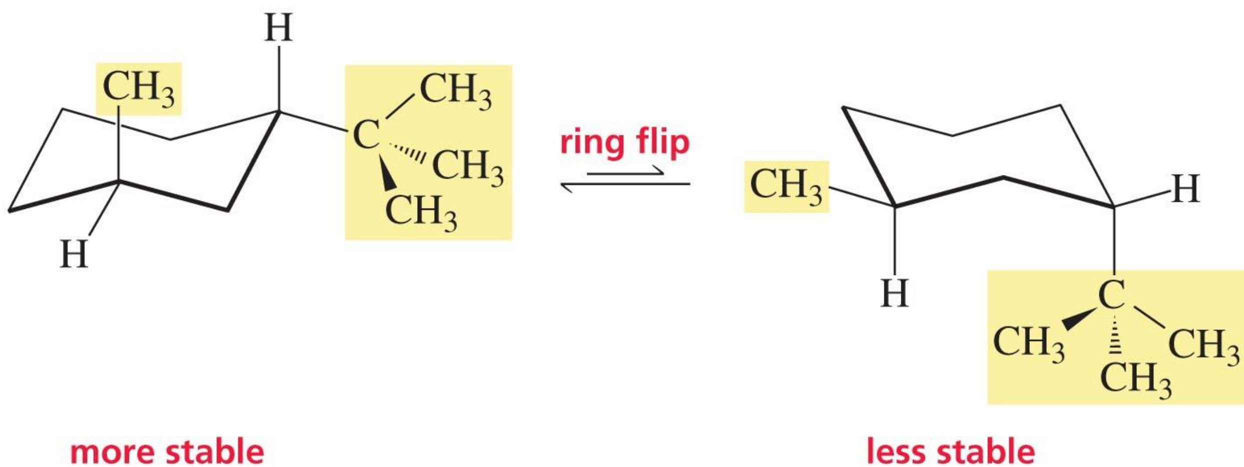


this chair conformer has four 1,3-diaxial interactions

1-*tert*-Butyl-3-Methylcyclohexane



cis-1-*tert*-butyl-3-methylcyclohexane



trans-1-*tert*-butyl-3-methylcyclohexane

Q: Which compound has the greatest torsional strain in the planar conformation?

A) Cyclopropane
B) Cyclobutane
C) Cyclopentane
D) cyclohexane

Q: Which compound has the greatest torsional strain in the planar conformation?

A) Cyclopropane

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C) Cyclopentane

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Q: The most stable conformation of *cis*-1-tert-butyl-4-methylcyclohexane has

A) both groups equatorial

B) both groups axial

C) the *tert*-butyl group axial and the methyl equatorial

D) the *tert*-butyl group equatorial and the methyl axial

Q: The most stable conformation of *cis*-1-tert-butyl-4-methylcyclohexane has

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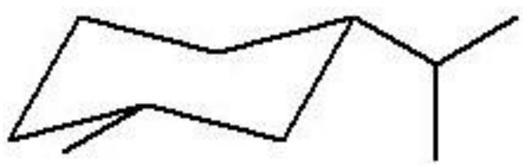
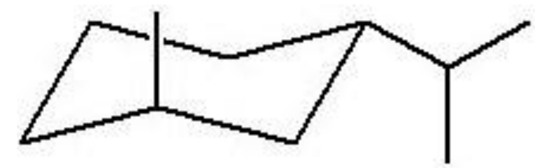
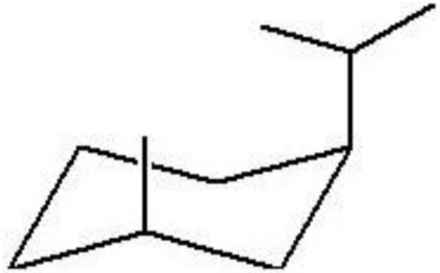
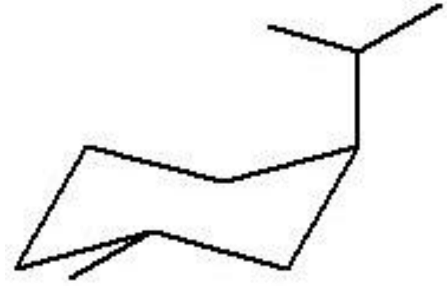
D) the *tert*-butyl group equatorial and the methyl axial

Q: The most stable conformation of *trans*-1-isopropyl-3-methylcyclohexane is:

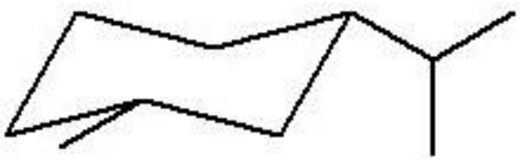
A) 

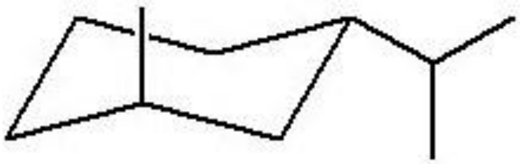
B) 

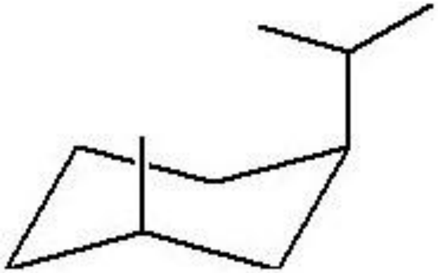
C) 

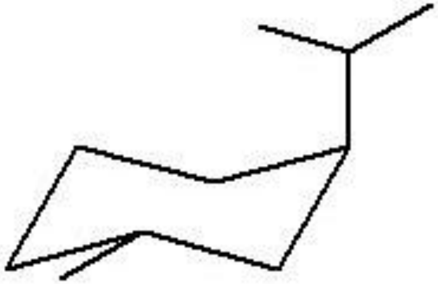
D) 

Q: The most stable conformation of *trans*-1-isopropyl-3-methylcyclohexane is:

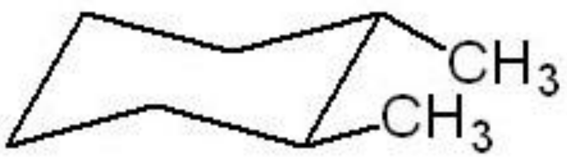
A) 

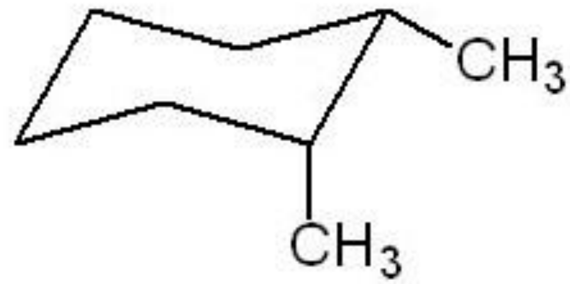
B) 

C) 

D) 

Q: What is the relationship between the two cyclohexane chairs below?





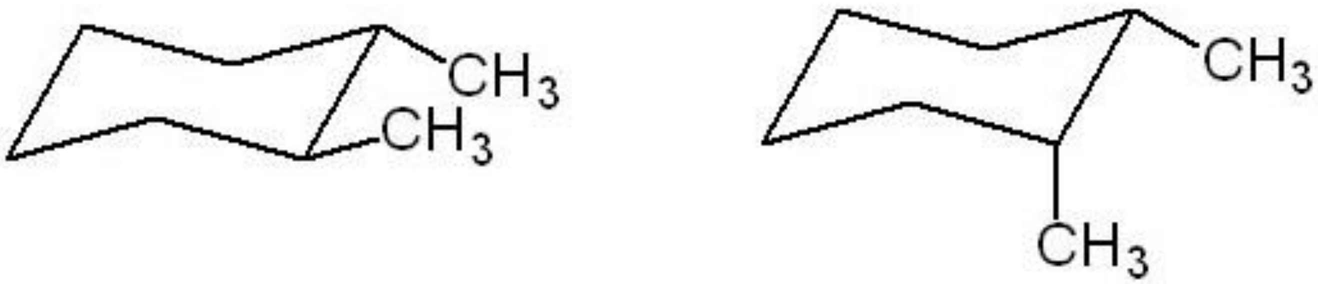
A) conformational isomers

B) constitutional isomers

C) Stereoisomers

D) different compounds

Q: What is the relationship between the two cyclohexane chairs below?



A) conformational isomers

B) constitutional isomers

C) Stereoisomers

D) different compounds

Solubility

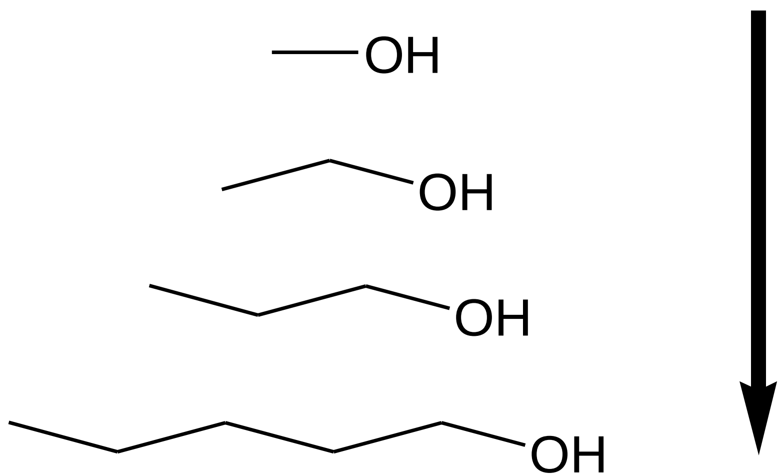
Alcohols:

R—OH

Nonpolar
Van der waals forces

Polar
Hydrogen bonds

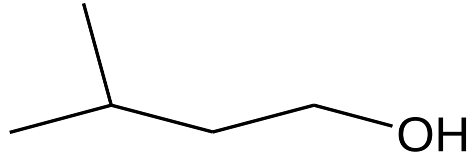
As the size of nonpolar part increase the water solubility of alcohol decrease



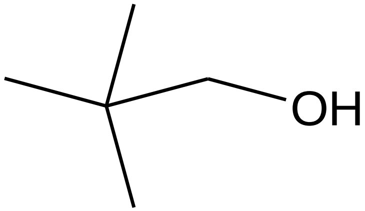
Solubility decrease

Solubility

1

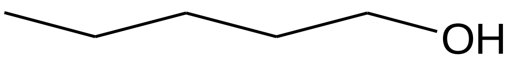


2

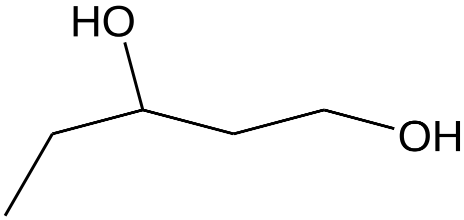


3

Solubility increase



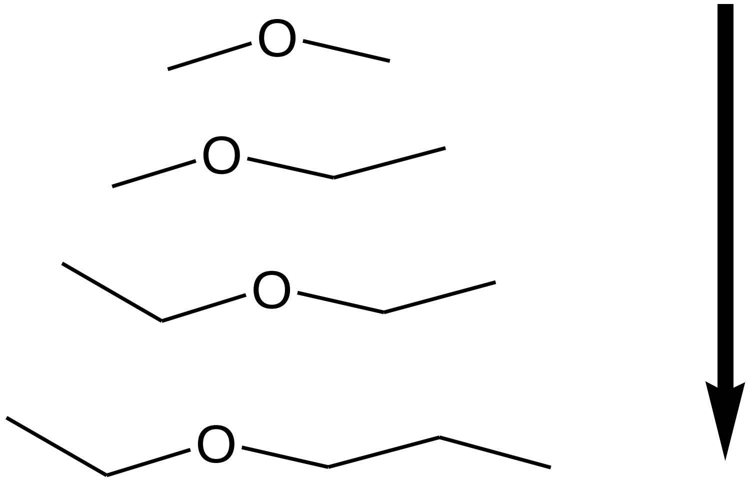
1



2

2 is more soluble

Ether solubility



Solubility decrease